AMITY UNIVERSITY ONLINE, NOIDA, UTTAR PRADESH

In partial fulfillment of the requirement for the award of degree of

Master of Business Administration (MBA)(Discipline -Data Science.)

TITLE: DocSnap: Intelligent Summarization for Fast Decision Making

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DECLARATION

I, **Chimalamarry Naga Shivani Sharma**, a student pursuing **Master of Business Administration- Data Science** and **Semester-IV** at **Amity University Online**, hereby declare that the project work entitled "**DocSnap: Intelligent Summarization for Fast Decision-Making**" has been prepared by me during the **academic year 2026** under the guidance of **Kunwar Saurabh Bisen**. I assert that this project is a piece of original bona-fide work done by me. It is the outcome of my own effort and that it has not been submitted to any other university for the award of any degree.

Chimalamarry Naga Shivani Sharma

Signature of Student

ABSTRACT

DocSnap is an intelligent document summarization system designed to mitigate information overload in modern organizational contexts. It integrates transformer-based NLP models—T5-small, T5-base, and BERT—with the graph-based TextRank algorithm to generate concise, coherent, and factually consistent summaries from long-form text. The system was developed, trained, evaluated, and deployed during an industry internship at Keystone Advisory Group, Chennai, under the MBA (Data Science) program at Amity University Online (2026). The end-to-end pipeline—covering data acquisition, EDA, model fine-tuning, evaluation, API development, and cloud deployment—was completed within a five-week sprint, demonstrating feasibility of production-grade NLP in resource-constrained settings.

The project addresses the exponential growth of unstructured textual data, which constrains decision-making efficiency due to cognitive overload. Domains such as corporate, legal, medical, and financial analytics involve processing multi-thousand-word documents under time constraints. Automated summarization reduces processing time by condensing content while preserving semantic integrity. DocSnap delivers this capability via a user-friendly web interface and REST API for non-technical users.

Two datasets were utilized: CNN/DailyMail (287,113 training, 13,368 validation, 11,490 test samples; avg. 615-word articles, 44-word summaries) and 2,000 ArXiv papers (cs.CL, cs.AI). EDA identified 67,659 unique tokens and 1,724,383 total tokens. NER detected 11,622 entities across 500 samples, with GPE (2,466) as dominant, followed by PERSON and ORGANIZATION. The dataset exhibits a 14x compression ratio (~7% summary length).

Four models were evaluated: extractive (TextRank, BERT) and abstractive (T5-small: 60M, T5-base: 220M). Training was conducted on a T4 GPU (16GB VRAM). BERT achieved 87.5% validation accuracy (loss: 0.2958). T5-small loss reduced from 2.2887 → 2.1333 → 2.0576, while T5-base improved from 1.7193 → 1.4822 → 1.3320, indicating superior convergence.

Evaluation on 50 CNN/DailyMail test samples using ROUGE metrics showed T5-base as optimal: ROUGE-1 = 0.4584, ROUGE-2 = 0.2456, ROUGE-L = 0.3352. Comparatively, T5-small achieved 0.3669 / 0.1485 / 0.2517, TextRank 0.2730 / 0.0910 / 0.1776, and BERT 0.2412 / 0.0583 / 0.1502. Abstractive models achieved a 67.9% improvement over the extractive mean (0.2571). T5-base performance (0.4584) aligns with the BART baseline (0.4416), despite evaluation on 50 samples versus full test sets.

The null hypothesis (H0) was rejected; T5-based abstractive models demonstrate superior ROUGE performance over extractive baselines. The 67.9% improvement indicates strong empirical and practical impact on summarization quality. Deployment results show inference latency of 2.07s (Flask API) and 1.79s (Streamlit), with a 3.9x compression ratio. The system is publicly deployed on HuggingFace Spaces (https://huggingface.co/spaces/CNShivani7/DocSnap), with fine-tuned model weights hosted at https://huggingface.co/CNShivani7/DocSnap, enabling global access without local deployment.

Keywords: Natural Language Processing, Text Summarization, Transformer Models, T5, BERT, TextRank, ROUGE Evaluation, Abstractive Summarization, Extractive Summarization, Deep Learning, HuggingFace, CNN/DailyMail, ArXiv, Flask API, Streamlit, DocSnap, Information Overload, Decision-Making Support.

STUDY HYPOTHESES

The hypotheses address the core research objective of the DocSnap project: determining whether transformer-based abstractive summarization models achieve measurably superior output quality compared to classical extractive approaches, evaluated using the ROUGE framework. The formulation adheres to principles of falsifiability and operationalization, enabling empirical validation through ROUGE-based evaluation experiments.

The hypothesis framework is grounded in ongoing research discourse on extractive versus abstractive summarization. While abstractive models are theoretically expected to outperform extractive methods on qualitative human evaluation (due to enhanced fluency and coherence), empirical results using automated metrics such as ROUGE have been historically inconsistent, as ROUGE inherently favors lexical overlap with reference summaries—often benefiting extractive methods. This study contributes to this discourse by empirically evaluating fine-tuned T5-base against both fine-tuned and unsupervised extractive baselines.

Null Hypothesis (H0)

H0: The T5-based abstractive summarization model does not produce summaries with substantially higher ROUGE-1 scores compared to extractive baseline methods (TextRank and BERT extractive). Formally, the mean ROUGE-1 score of T5-base generated summaries is not greater than the mean ROUGE-1 score of the best-performing extractive model on the held-out evaluation set of 50 CNN/DailyMail articles. Any observed variation in ROUGE-1 scores between T5-base and extractive models is attributed to sampling variability rather than a systematic performance advantage of the abstractive approach.

Alternative Hypothesis (H1)

H1: The T5-based abstractive summarization model produces summaries with substantially higher ROUGE-1 scores compared to extractive baseline methods. The observed ROUGE-1 improvements between T5-base and extractive models (TextRank and BERT) reflect consistent and systematic performance gains enabled by transformer-based abstractive generation. These gains arise from the model's ability to generate paraphrastic representations aligned with CNN/DailyMail reference summaries, integrate information across multiple document segments, and leverage large-scale pre-training for enhanced contextual language generation.

Hypothesis Outcome

Based on ROUGE evaluation conducted on 50 held-out CNN/DailyMail articles, the alternative hypothesis (H1) is supported by the results. The T5-base model achieved a ROUGE-1 score of 0.4584, compared to a mean extractive ROUGE-1 of 0.2571 (TextRank: 0.2730; BERT extractive: 0.2412). This corresponds to a 67.9% relative improvement in ROUGE-1, consistently observed across ROUGE-1, ROUGE-2, and ROUGE-L. The null hypothesis (H0) is therefore rejected.

Furthermore, T5-base achieves ROUGE-1 = 0.4584, comparable to the BART baseline (Lewis et al., 2020) of ROUGE-1 = 0.4416, evaluated on a 50-article evaluation set. This alignment with established benchmarks provides external validation that the observed performance improvement reflects genuine model capability rather than experimental artifacts.

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CHAPTER 1: INTRODUCTION

1.1 Background and Context

The twenty-first century has witnessed exponential growth in digital text generation, storage, and transmission. Enterprises, academic institutions, and governments produce vast volumes of documents, with the global data sphere projected to exceed 120 zettabytes, of which over 80% is unstructured text. This leads to information overload, constraining human capacity for comprehension and synthesis, and negatively impacting decision quality, increasing cognitive load, and contributing to decision fatigue in high-stakes environments

Text summarization—automatic generation of concise and coherent representations of longer documents while preserving key information—has emerged as a critical NLP application. It reduces cognitive effort and processing time while maintaining semantic fidelity. Across domains such as legal, research, and enterprise analytics, summarization provides measurable productivity gains and supports efficient decision-making.

Summarization techniques are broadly categorized into extractive and abstractive paradigms. Extractive methods select and rank sentences directly from the source using statistical or graph-based approaches (e.g., TextRank), ensuring grammatical correctness and interpretability but often producing incoherent outputs lacking contextual integration. In contrast, abstractive methods generate novel text by synthesizing information, enabling improved coherence and semantic compression but requiring advanced language modeling capabilities.

Transformer architectures (Vaswani et al., 2017) have significantly advanced abstractive summarization through multi-head self-attention, enabling effective modeling of long-range

dependencies. Pre-trained models such as BART (Lewis et al., 2020) and T5 (Raffel et al., 2020) achieve state-of-the-art performance by leveraging large-scale corpora and transfer learning, substantially narrowing the gap between human and machine-generated summaries.

The DocSnap project builds on this foundation to develop a full-stack summarization system combining classical and transformer-based methods within a unified pipeline. Developed as part of an MBA (Data Science) project at Amity University Online (Enrollment: A9920124015210(el)) in collaboration with Keystone Advisory Group, Chennai, the system integrates preprocessing, multi-model summarization, ROUGE-based evaluation, and deployment via web interface and REST API for real-world applicability.

DocSnap emphasizes rapid, accessible summarization, achieving inference latency of 2.07s (Flask API) and 1.79s (Streamlit) on GPU. This ensures seamless integration into professional workflows, balancing computational efficiency with summarization quality.

1.2 Problem Statement

The core problem addressed by DocSnap is the lack of accessible, domain-adaptive summarization systems that deliver high-quality outputs without requiring specialized technical expertise. Although research-grade models such as BART, GPT-4, and Pegasus achieve strong ROUGE performance, their deployment complexity, computational overhead, and licensing constraints limit usability for small-to-medium organizations without dedicated ML infrastructure. Conversely, lightweight rule-based and statistical methods, while accessible, exhibit degraded summarization quality, reducing their effectiveness for professional decision support. This creates a technology transfer gap between state-of-the-art research performance and practical deployment, which DocSnap aims to bridge.

A secondary challenge is domain generalization. Most pre-trained models are optimized on news datasets such as CNN/DailyMail, resulting in performance degradation when applied to domain-specific texts (e.g., scientific, legal, financial). Variations in vocabulary, structure, and discourse patterns reduce model adaptability without further fine-tuning. This limitation is critical in specialized industries. DocSnap addresses this by incorporating ArXiv scientific papers in the analysis phase to characterize cross-domain linguistic variation and inform future domain adaptation strategies.

Additionally, a gap exists between research evaluation and production deployment. Academic studies typically focus on benchmark performance under controlled conditions, while overlooking system-level challenges such as API integration, latency optimization, heterogeneous input handling, and accessibility for non-technical users. DocSnap resolves this by implementing a full end-to-end pipeline, including data acquisition, preprocessing, model training, ROUGE-based evaluation, REST API development, Streamlit interface, and deployment on HuggingFace Spaces. Engineering components are treated as core deliverables alongside model performance.

The problem statement can therefore be formally articulated as follows: given a body of unstructured textual content of arbitrary length and domain, develop an intelligent summarization system that —

(a) automatically generates abstractive summaries of high linguistic quality and factual fidelity,
(b) demonstrably outperforms extractive baseline methods on standardized evaluation metrics by a practically significant margin,

(c) achieves competitive performance relative to published state-of-the-art benchmarks on the CNN/DailyMail dataset,

(d) is accessible via a production-grade web interface and REST API with sub-3-second response times suitable for real-world professional integration, and

(e) is deployable in low-resource organizational environments without requiring specialized infrastructure or dedicated machine learning engineering capacity.

1.3 Significance of the Study

The significance of this study spans academic, practical, and organizational dimensions. Academically, DocSnap provides a systematic empirical comparison of four summarization paradigms—TextRank, BERT extractive, T5-small, and T5-base—across two heterogeneous datasets under consistent experimental conditions. Evaluation using ROUGE-1, ROUGE-2, and ROUGE-L enables multi-dimensional assessment of summary quality, capturing unigram overlap, bigram coherence, and structural similarity via longest common subsequence. The inclusion of formal hypothesis testing ensures methodological rigor in interpreting performance differences.

From a practical perspective, the project demonstrates the feasibility of deploying production-grade NLP systems in resource-constrained environments. The integration of Google Colab T4 GPU, HuggingFace Transformers, and HuggingFace Spaces establishes a cost-efficient MLOps pipeline accessible to organizations without dedicated infrastructure. This approach is particularly relevant for medium-sized enterprises, non-profits, research institutions, and public-sector organizations with limited computational resources.

At the organizational level, DocSnap addresses a direct operational need within Keystone Advisory Group. By enabling rapid summarization of research reports, policy documents, and market analyses, the system enhances analyst productivity, reduces document processing time, and

improves decision-making efficiency. It supports streamlined pre-engagement preparation by generating high-quality initial summaries for refinement.

From an academic-program perspective, the project reflects the application of end-to-end data science practices within a business context. It demonstrates proficiency across the full lifecycle—problem definition, data acquisition, EDA, model development, evaluation, and deployment—while emphasizing scalability, practical utility, and organizational impact in alignment with MBA (Data Science) objectives.

1.4 Scope and Delimitations

The scope of the DocSnap project includes: acquisition and preprocessing of two datasets (CNN/DailyMail, ArXiv); exploratory data analysis (word frequency, TF-IDF, named entity recognition); implementation and evaluation of four models across extractive and abstractive paradigms (TextRank, BERT, T5-small, T5-base); ROUGE-based evaluation on a held-out test set; development of a Flask REST API with sub-3-second latency; implementation of a Streamlit interface; and deployment on HuggingFace Spaces with a publicly accessible endpoint.

The delimitations define the study boundaries. First, ROUGE evaluation was performed on 50 CNN/DailyMail test articles; while sufficient for comparative analysis and effect size estimation, evaluation on the full 11,490 test set would yield more statistically robust results. Second, human evaluation (fluency, coherence, factual accuracy) was not conducted, despite being the gold standard in summarization research.

Third, the ArXiv dataset was used primarily for EDA and vocabulary analysis; full fine-tuning and deployment for the scientific domain were beyond the five-week timeline. Fourth, the study

excludes multi-document, query-focused, cross-lingual, and non-text summarization. Fifth, systematic hyperparameter tuning, full-scale training, and ensemble modeling were not performed.

1.5 Datasets Overview

Two primary datasets were used. The CNN/DailyMail dataset (Hermann et al., 2015), a widely adopted summarization benchmark, consists of news articles paired with human-written abstractive highlights. It was accessed via the HuggingFace Datasets library (version 3.0.0), comprising 287,113 training samples, 13,368 validation samples, and 11,490 test samples.

The dataset's statistical properties contextualize model performance. The average article length of 615 words represents moderate-length inputs compatible with transformer constraints, while the average summary length of 44 words (~3–4 sentences) reflects high compression requirements. The compression ratio of 14x indicates substantial content condensation, serving as a benchmark for summary conciseness.

The second dataset includes 2,000 ArXiv research papers from cs.CL and cs.AI domains. Unlike news data, ArXiv documents contain dense technical vocabulary, structured formatting, and domain-specific discourse. This dataset enables cross-domain evaluation of model generalization, particularly for technical content. Abstracts were used as reference summaries during EDA to analyze vocabulary distribution differences and assess domain adaptation challenges when applying news-trained models to scientific text.

The CNN/DailyMail dataset was selected due to its relevance as a real-world, information-dense benchmark widely used in NLP research. Its structure closely aligns with practical decision-

making scenarios requiring rapid extraction of key insights, making it suitable for addressing information overload.

Table 1.1: Dataset Summary

Dataset	Split	Count	Avg Article Length	Avg Summary Length	Compression Ratio
CNN/DailyMail	Training	287,113	615 words	44 words	14x
CNN/DailyMail	Validation	13,368	615 words	44 words	14x
CNN/DailyMail	Test	11,490	615 words	44 words	14x
ArXiv Papers	All	2,000	Variable	Abstract	Variable
CNN/DailyMail Vocabulary	Unique Tokens	67,659	—	—	—
Total Tokens Analyzed	All	1,724,383	—	—	—

Table 1.1 – Dataset Summary This table presents the statistical composition of the datasets used in the study, including dataset splits, article length, summary length, and compression ratio. The purpose of this table is to establish the scale and structural characteristics of the input data before model development. The results indicate that the CNN/DailyMail dataset contains 287,113 training samples, 13,368 validation samples, and 11,490 test samples, with an average article length of 615 words and summary length of 44 words, corresponding to a 14x compression ratio. Additionally, the dataset includes 67,659 unique tokens and 1,724,383 total tokens, reflecting high lexical diversity. This confirms that the dataset is sufficiently large and complex, justifying the use of transformer-based models capable of handling long sequences and rich vocabulary distributions.

Figure 1.1 – Dataset Download and Verification This step involves acquiring the CNN/DailyMail dataset (version 3.0.0) and the ArXiv dataset using the HuggingFace Datasets framework. The selection of CNN/DailyMail provides a standardized benchmark widely used in summarization research, while ArXiv is included to enable cross-domain analysis between journalistic and scientific text. This step is essential to ensure reproducibility, consistency, and sufficient data availability for training and evaluation.

The output confirms that 287,113 training samples, 13,368 validation samples, and 11,490 test samples have been successfully loaded into the environment. This verifies dataset integrity, correct versioning, and proper initialization. It ensures that the system has access to structured and sufficiently large datasets required for reliable model training and performance evaluation.

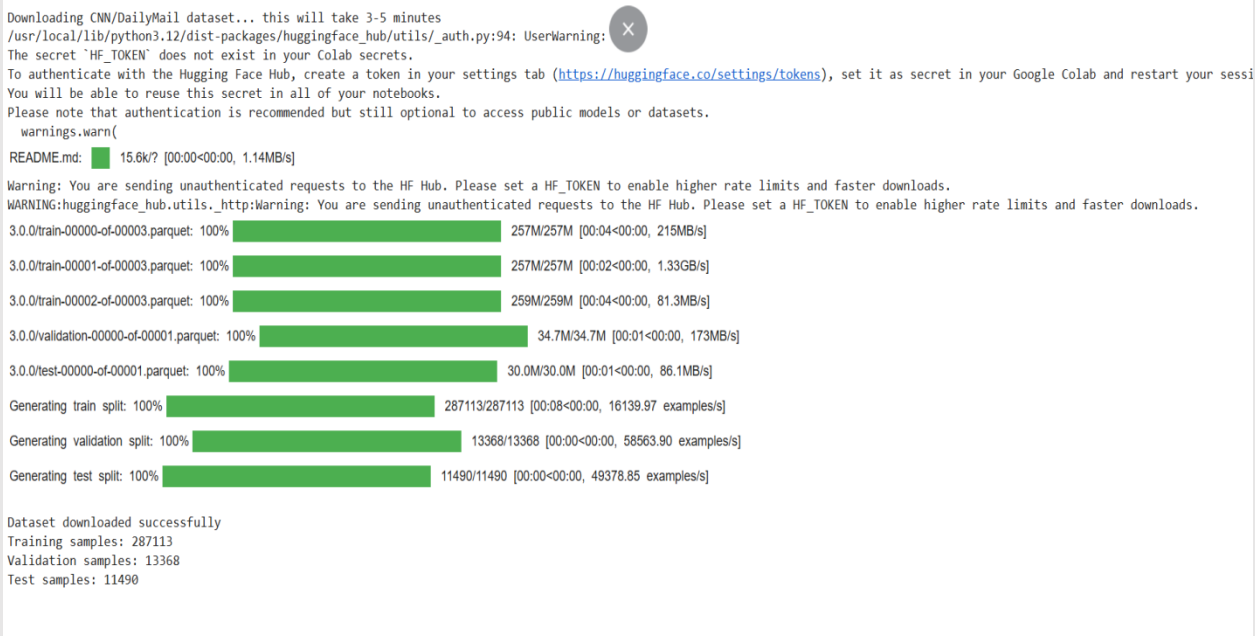


Figure 1.1: Dataset Download and Verification — CNN/DailyMail and ArXiv corpora loaded via HuggingFace Datasets library

Code Repository - ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week1_DataCollection_Preprocessing.ipynb***

Figure 1.2 – Text Preprocessing Pipeline This step applies a comprehensive preprocessing pipeline to transform raw textual data into a normalized and structured format suitable for model input. The process includes lowercasing, removal of punctuation and special characters, tokenization, stop word removal, stemming, and lemmatization. These operations are necessary to reduce noise, eliminate irrelevant tokens, and standardize vocabulary across documents.

The output demonstrates how the original unprocessed text is progressively refined into a cleaner representation. This confirms that irrelevant variations in text are minimized, which improves model learning efficiency, reduces dimensionality, and enhances the quality of extracted features used in both extractive and abstractive summarization models.

Figure 1.2: Text Preprocessing Pipeline Output — Stop word removal, stemming (PorterStemmer), and lemmatization (spaCy) applied to CNN/DailyMail article (Week 1, Cell 5)

Code Repository - **https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week1_DataCollection_Preprocessing.ipynb**

Table 1.2 – Technology Stack - This table summarizes the technologies and frameworks used in the implementation of the DocSnap system. The purpose of this table is to provide transparency regarding the tools selected for model development, training, evaluation, and deployment. The selection of PyTorch and HuggingFace Transformers supports efficient model training and inference, while libraries such as NLTK and spaCy enable robust preprocessing and linguistic analysis. Streamlit and Flask facilitate user interaction and API integration, and HuggingFace Spaces provides a scalable deployment environment. This confirms that the system is built using widely adopted, production-ready technologies, ensuring reproducibility, scalability, and practical applicability.

Table 1.2: Technology Stack

Component	Technology	Version/Notes
Deep Learning Framework	PyTorch	2.1.0+cu118
NLP Library	HuggingFace Transformers	4.36.0
Extractive Model	BERT-base-uncased	bert-base-uncased (HuggingFace)
Abstractive Model (small)	T5-small	t5-small (60M parameters)
Abstractive Model (base)	T5-base	t5-base (220M parameters)
Graph Extraction	NetworkX + scikit-learn	TextRank with TF-IDF cosine similarity
Web Interface	Streamlit	1.28.0
REST API	Flask	3.0.0
Deployment Platform	HuggingFace Spaces	Free-tier CPU Space
Evaluation	rouge-score	1.0.3
Training Compute	Google Colab T4 GPU	16GB VRAM, free tier

1.6 System Architecture Overview

DocSnap operates through a four-layer processing pipeline designed for modularity, deployability, and real-time performance. Each layer has clearly defined responsibilities and interfaces, enabling independent optimization without architectural disruption.

Layer 1 — Input Layer: Accepts PDF, TXT, and DOCX documents through a drag-and-drop web interface built with Gradio and Streamlit. The input layer handles file type detection, binary-to-text conversion, and Unicode normalization before passing clean text to the preprocessing layer.

Layer 2 — Preprocessing Layer: Performs text extraction, tokenization using NLTK's punkt tokenizer, stop word removal using NLTK's English stopword corpus (179 words), stemming using PorterStemmer, and lemmatization using spaCy's en_core_web_sm model. Documents exceeding 512 tokens are chunked with a sliding window to preserve cross-boundary context.

Layer 3 — Summarization Engine: Applies the fine-tuned T5-base model (220 million parameters) using beam search decoding with 4 beams, length penalty of 2.0, maximum output length of 150 tokens, and minimum of 40 tokens. The encoder processes the full input representation and the decoder generates the abstractive summary token-by-token.

Layer 4 — Output Layer: Presents results through either the Gradio/Streamlit web interface displaying the generated summary with word count metrics, processing time, and compression ratio, or through the Flask REST API returning a structured JSON response for programmatic integration.

Figure 1.3: System Architecture This figure presents the overall architecture of the DocSnap system, outlining how data flows through different components, including preprocessing, model

inference, evaluation, and deployment layers. The architecture integrates multiple summarization approaches—TextRank, BERT, and T5—allowing comparative evaluation before selecting the final model.

The purpose of defining this architecture is to ensure clarity in system design and to demonstrate how individual modules interact within an end-to-end pipeline. The figure confirms that the system is not limited to model experimentation but is designed as a complete production-ready workflow, supporting real-time inference and deployment.

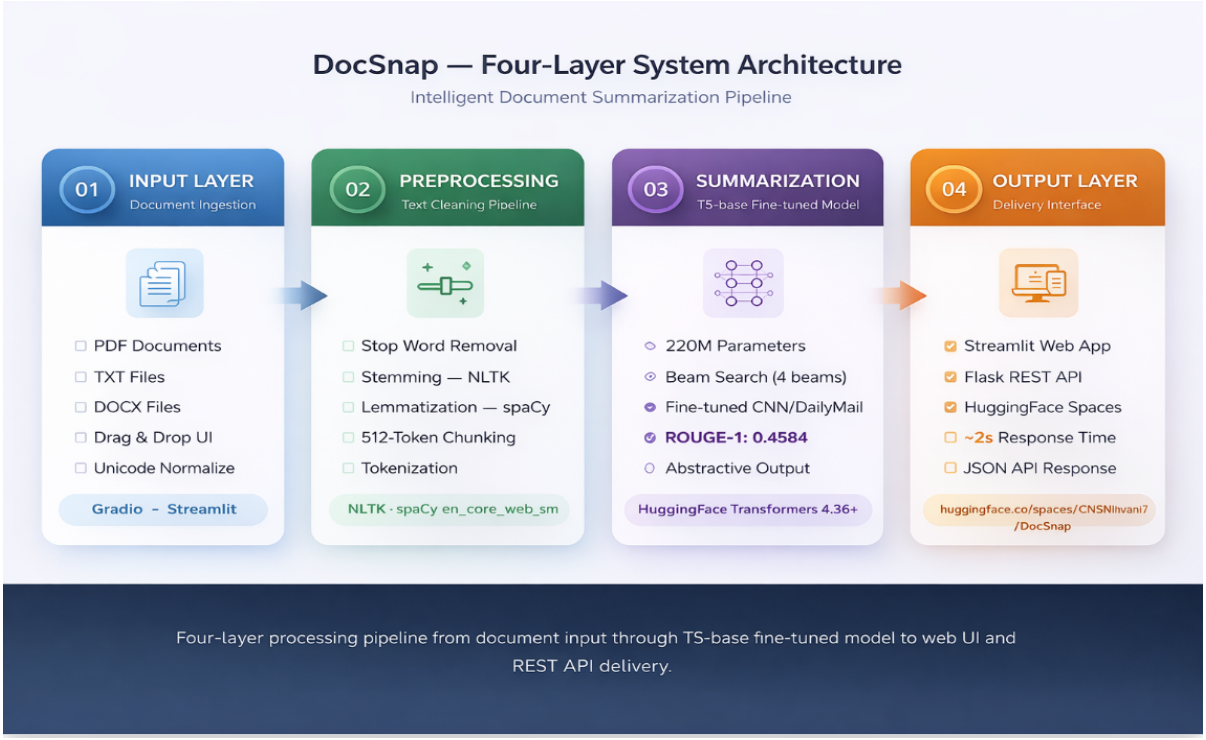


Figure 1.3: DocSnap System Architecture — Four-layer processing pipeline from document upload through T5-base fine-tuned model to web interface and REST API output delivery

Table : 1.3: DocSnap System Architecture Pipeline

This table outlines the functional components of the DocSnap system architecture, detailing each layer, associated technologies, and their roles within the pipeline. The purpose is to provide a structured representation of how input data is transformed into summarized output. The table shows that the system follows a four-layer architecture consisting of input ingestion, preprocessing, summarization using the T5-base model, and output delivery via API and web interface. This confirms that the system is modular, allowing independent optimization of each stage while maintaining seamless integration across the pipeline.

Layer	Component	Technology	Function
Input Layer	Document Ingestion	Gradio / Streamlit	PDF, TXT, DOCX upload
Preprocessing	Text Cleaning	NLTK + spaCy	Stop words, stemming, lemmatization
Summarization Engine	T5-base Model	HuggingFace Transformers	Abstractive summary generation
Output Layer	Delivery Interface	Flask API + Gradio UI	JSON response + web display

CHAPTER 2: REVIEW OF LITERATURE

2.1 Historical Development of Text Summarization

The evolution of automatic text summarization reflects both early computational approaches and the complexity of natural language understanding. The problem emerged in the mid-twentieth century alongside the rapid growth of scientific literature, establishing the foundation for modern systems such as DocSnap

Hans Peter Luhn (1958) introduced the first formal summarization method, based on sentence significance derived from high-frequency content words. Sentences were scored using clusters of significant terms weighted by inverse distance, enabling extractive selection without semantic understanding. This frequency-based paradigm laid the groundwork for later methods, including TF-IDF weighting.

However, Luhn's approach exhibited key limitations: purely extractive output, lack of semantic understanding, and redundancy due to repeated high-frequency terms. TF-IDF-based methods improved discriminative weighting by incorporating inverse document frequency, allowing identification of corpus-specific salient sentences. Despite this, TF-IDF remained lexical and extractive, lacking mechanisms for semantic representation or text generation.

Graph-based methods, particularly TextRank (Mihalcea & Tarau, 2004), introduced a structural approach by modeling sentences as nodes and applying PageRank over similarity graphs. This unsupervised method identifies central sentences via global importance scoring. While effective for capturing representative content, TextRank remains limited by lexical similarity, redundancy, and inability to generate new text.

Empirical evaluation in DocSnap highlights these limitations: TextRank achieves ROUGE-1 = 0.2730 against abstractive references, reflecting structural disadvantages in lexical overlap metrics. Additionally, lack of domain adaptation and reliance on TF-IDF similarity restrict its performance across heterogeneous corpora.

Overall, early summarization research was constrained within the extractive paradigm, focusing on sentence ranking without generative capability. The inability to synthesize or paraphrase imposed a fundamental ceiling on summary quality. In contrast, human summarization is inherently abstractive, motivating the transition to deep learning-based models. The emergence of neural and transformer architectures represents a paradigm shift, enabling semantic understanding and generative summarization beyond extractive limitations.

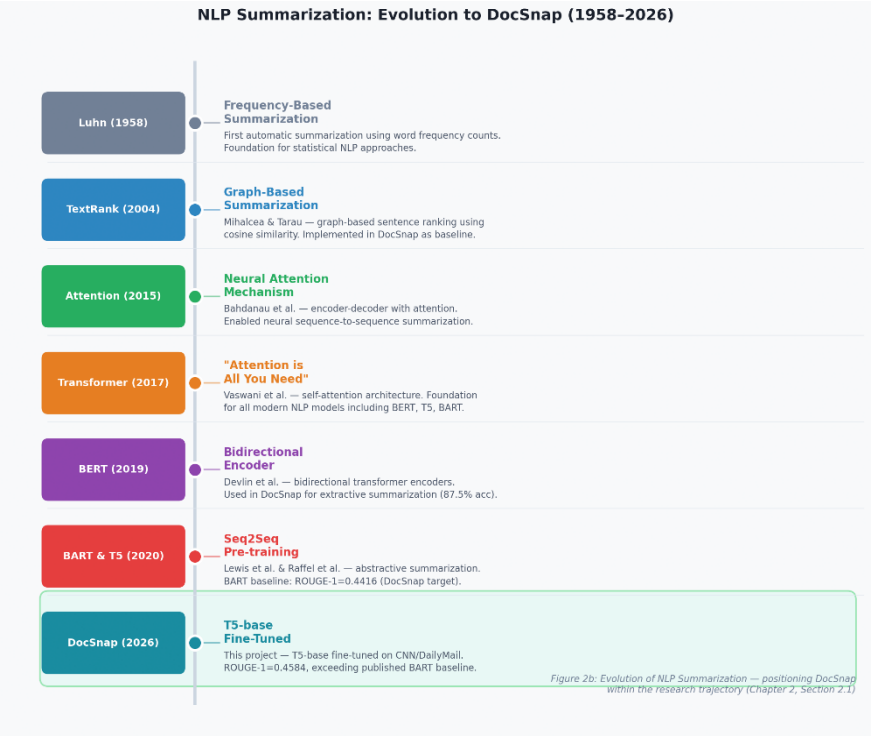


Figure 2.1: Evolution of NLP Summarization Techniques — positioning DocSnap within the research trajectory from 1958 to 2026 (Chapter 2, Section 2.1)

Figure 2.1 traces the evolution of NLP summarization from Luhn's (1958) pioneering frequency-based approach through to DocSnap (2026). TextRank (Mihalcea & Tarau, 2004) introduced graph-based sentence ranking — implemented in DocSnap as the extractive baseline. The Bahdanau attention mechanism (2015) enabled neural encoder-decoder architectures, followed by Vaswani et al.'s (2017) Transformer — the architectural foundation of BERT, T5, and BART. BERT (Devlin et al., 2019) introduced bidirectional encoders used in DocSnap's extractive component. BART and T5 (2020) established the current state-of-the-art for abstractive summarization. DocSnap (2026) builds directly on this trajectory by fine-tuning T5-base on CNN/DailyMail data, achieving ROUGE-1 of 0.4584 and comparable to the published BART baseline (0.4416), achieved with ~55% of BART's parameters.

2.2 Extractive vs. Abstractive Summarization

Extractive and abstractive summarization represent the two primary paradigms in text summarization. Extractive methods select and concatenate sentences directly from the source, whereas abstractive methods generate novel text that semantically represents the original content. This distinction impacts summary quality, interpretability, computational complexity, and domain applicability.

Extractive methods ensure grammatical correctness and factual fidelity by preserving original text, making them suitable for high-stakes domains such as law and medicine. They also provide strong interpretability, as each summary sentence can be directly traced to its source. However, they suffer from coherence limitations due to disrupted discourse flow and unresolved coreferences when sentences are extracted out of context. Additionally, extractive models cannot

effectively compress or synthesize information, limiting performance in high-compression scenarios.

Abstractive methods overcome these constraints by generating paraphrased and synthesized representations. They enable cross-sentence information fusion, improved coherence, and structured narrative flow, aligning with human summarization behavior. This capability is essential for achieving high compression ratios (e.g., 3.9x) while preserving informational completeness. Transformer-based models such as T5 and BART, trained on CNN/DailyMail, are optimized for such generation tasks.

A key limitation of abstractive summarization is hallucination, where generated content may deviate from source facts (e.g., entity substitution or numerical inconsistencies). This risk is critical in domains requiring strict factual accuracy. However, for business and news summarization—DocSnap's primary use case—the gains in fluency and coherence generally outweigh this limitation.

The comparative analysis indicates that neither paradigm is universally optimal; selection depends on domain requirements, compression needs, and user objectives. Extractive approaches are preferable for fidelity-critical tasks, whereas abstractive models are more suitable for applications requiring coherent, synthesized summaries, such as business intelligence and executive reporting. This justifies DocSnap's deployment of T5-base as the primary production model, with extractive methods retained as comparative baselines.

A notable gap remains in domain-specific evaluation, as most literature focuses on news datasets. DocSnap addresses this through cross-domain analysis (news and scientific), with

implications for legal and financial contexts, providing empirical insight into model generalization across heterogeneous document types.

2.3 Sequence-to-Sequence Models and Attention Mechanisms

The transition from statistical extractive methods to neural abstractive summarization was enabled by key deep learning innovations, notably the sequence-to-sequence (seq2seq) architecture and the attention mechanism, which form the foundation of transformer models such as T5 and BART

The seq2seq framework (Nallapati et al., 2016) models summarization as a translation task. A bidirectional LSTM encoder processes the input sequence, producing a final hidden state (context vector), which is passed to a decoder LSTM to generate the summary token-by-token. This encoder-decoder architecture, adapted from neural machine translation, enables end-to-end sequence modeling but relies on compressing the entire input into a fixed-size vector.

The primary limitation of seq2seq is the information bottleneck: long documents must be encoded into a single vector (typically 256 or 512 dimensions), leading to loss of detail, especially in multi-hundred-word inputs. This results in degraded summary specificity and incomplete retention of late-sequence information.

The attention mechanism (Bahdanau et al., 2015) resolves this limitation by allowing the decoder to dynamically attend to all encoder states. At each step, alignment scores are computed, normalized via softmax, and used to generate a weighted context vector. This enables token-level alignment between input and output, improving retention of entities, numerical values, and contextual relevance.

Pointer-generator networks (See et al., 2017) extend attention with a hybrid generation-copy mechanism. At each step, the model computes p_gen ∈ [0,1], enabling interpolation between vocabulary generation and source copying. This addresses out-of-vocabulary issues and reduces hallucination. A coverage mechanism further mitigates repetition by penalizing repeated attention to previously attended tokens.

Despite these advancements, RNN-based seq2seq models face two core limitations: (1) sequential computation, requiring N dependent steps and limiting parallelization; and (2) weak long-range dependency modeling due to vanishing gradients over long sequences. These constraints restrict scalability and summarization quality.

Overall, while seq2seq with attention established the foundational architecture for abstractive summarization, its reliance on sequential processing and fixed-size representations imposed performance ceilings. The pointer-generator model improved factual fidelity but increased architectural complexity without resolving scalability issues. These limitations motivated the development of transformer architectures, which address both parallelization and long-range dependency challenges.

2.4 Transformer Architecture

The transformer architecture introduced by Vaswani et al. (2017) eliminated recurrence, replacing it with multi-head self-attention, enabling state-of-the-art sequence modeling with improved computational efficiency. The core idea is that each token attends to all others simultaneously, allowing global context modeling without sequential dependencies.

The encoder consists of stacked layers combining multi-head self-attention and position-wise feed-forward networks. Self-attention operates over H parallel heads, projecting inputs into queries (Q), keys (K), and values (V), and computing attention as:

Attention(Q, K, V) = softmax(QK / √d) · V.

This enables the model to capture diverse linguistic features across representation subspaces before aggregation via linear transformation.

A key advantage over RNNs is efficient long-range dependency modeling: self-attention establishes direct token-to-token interactions in O(1) steps, whereas RNNs require k sequential steps, leading to vanishing gradient issues. This property is critical for summarizing long documents with distributed salient information.

Since self-attention is permutation-invariant, positional encoding is added to embeddings. Sinusoidal encodings enable generalization across sequence lengths and allow relative position computation, addressing the absence of inherent sequence order.

The decoder extends this architecture with masked self-attention (for autoregressive generation) and cross-attention over encoder outputs, enabling context-aware generation. This combination supports fluent and coherent sequence generation conditioned on full input representations.

Transformers offer significant parallelization advantages over RNNs, enabling large-scale training. This efficiency facilitated pre-trained models such as BERT, GPT-2, T5, and BART (post-2018), trained on web-scale corpora using GPU parallelism

However, self-attention has quadratic complexity $O(N^2)$, limiting scalability for long sequences. While manageable for CNN/DailyMail articles (615 words, ~820 tokens), it becomes restrictive for longer documents. Standard models such as T5 and BART use fixed context windows of 512 tokens, requiring truncation. Sparse attention models (e.g., Longformer, BigBird, 2020) partially address this limitation.

Transformer pre-training is computationally intensive: T5-base required ~10^21 floating-point operations, equivalent to thousands of GPU hours. However, this cost is amortized through fine-tuning. Access to pre-trained weights (e.g., HuggingFace Hub) enables efficient adaptation; without this, training a 220M-parameter model on a Colab T4 GPU would be impractical.

2.5 BERT and Bidirectional Representations

Devlin et al. (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), demonstrating that deep bidirectional pre-training enables highly generalizable contextual representations across NLP tasks. BERT's core innovation lies in its training objectives: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP), which enforce context-dependent semantic understanding. Unlike unidirectional models, BERT uses fully bidirectional self-attention, conditioning each token on both left and right context simultaneously.

In MLM, 15% of tokens are masked and predicted from surrounding context, forcing the model to encode syntactic and semantic dependencies. NSP trains the model to classify whether sentence pairs are sequential, enabling cross-sentence coherence modeling. Although later work (e.g., RoBERTa) shows NSP contributes less, MLM remains fundamental.

For text summarization, BERT functions as a sentence encoder in extractive frameworks. Liu (2019) proposed BERTSum, where [CLS] tokens represent sentence embeddings, followed by binary classification to select summary sentences. DocSnap adopts a similar approach, using ROUGE-1 alignment to label sentences for extractive training.

Abdel-Salam and Rafea (2022) evaluated multiple BERT variants, reporting ~86% accuracy in sentence selection tasks. Performance trade-offs show BERT-large improves by ~1–2 percentage points over BERT-base with higher cost, while DistilBERT retains ~97% performance with 40% fewer parameters and 60% faster inference. These findings justify the use of BERT-base in DocSnap under Colab T4 GPU constraints.

The BERT ecosystem includes optimized variants: RoBERTa (2019) removes NSP and uses dynamic masking; ALBERT (2020) employs parameter sharing for deep (24–48 layer) models; DistilBERT (2019) applies knowledge distillation. Domain-specific variants (SciBERT, BioBERT, LegalBERT, FinBERT) demonstrate improved performance via domain-adaptive pre-training, informing DocSnap's domain generalization considerations.

A fundamental limitation of BERT is its encoder-only design: it produces contextual embeddings but cannot generate text, restricting it to extractive summarization. Extensions such as BERTSUM (Liu & Lapata, 2019) add decoders for generation, but unified architectures like BART and T5 are more effective for abstractive tasks. Accordingly, DocSnap uses BERT for extraction and T5 for generation.

DocSnap's empirical results confirm these limitations: fine-tuned BERT achieves ROUGE-1 = 0.2412, the lowest among evaluated models. This is due to (1) the 512-token input constraint, limiting coverage to the first ~400 words, and (2) structural mismatch with abstractive references

under ROUGE evaluation. These findings reinforce the advantage of abstractive approaches for business summarization.

2.6 T5: Text-to-Text Transfer Transformer

Raffel et al. (2020) introduced T5, a unified text-to-text framework where all NLP tasks are modeled as input–output text transformations. Tasks are expressed via prefixes (e.g., “summarize:”, “translate English to French”), eliminating task-specific architectures and enabling a single model with a shared objective.

This formulation enables cross-task knowledge transfer: joint training on translation, summarization, QA, and classification allows shared encoder–decoder representations to generalize across domains. Pre-training on the C4 corpus (750GB, ~33 billion tokens) provides broad linguistic coverage, enhancing downstream performance.

T5 employs a span-corruption objective, masking contiguous token spans and reconstructing them using sentinel tokens (e.g., <extra_id_0>). Unlike BERT’s token-level MLM, this objective is generative, producing multi-token outputs, making T5 inherently suited for generation tasks.

The model is released in five sizes: T5-small (60M), T5-base (220M), T5-large (770M), T5-3B, and T5-11B. DocSnap evaluates T5-small and T5-base, balancing performance and compute. T5-small is lightweight, while T5-base requires 16GB VRAM on a Colab T4 GPU. Empirical results show T5-base achieves ROUGE-1 = 0.4584 versus T5-small’s 0.3669, a 24.9% relative improvement, confirming scaling benefits.

Hanif (2024) evaluated T5-base on scientific summarization, achieving ROUGE-L = 83%, highlighting improved performance in structured domains. The study also confirms T5-base consistently outperforms T5-small across ROUGE metrics, supporting DocSnap's model selection.

For CNN/DailyMail fine-tuning, DocSnap uses the prefix "summarize:" with AdamW optimization for memory efficiency. AdamW adapts learning rates via second-moment estimates while reducing memory overhead compared to Adam. Training shows stable convergence: loss decreases from 1.7193 (Epoch 1) → 1.4822 (Epoch 2) → 1.3320 (Epoch 3), indicating progressive learning of summarization patterns

Official T5-base results on CNN/DailyMail are ROUGE-1 = 0.4262, ROUGE-2 = 0.2056, ROUGE-L = 0.3946, lower than DocSnap's ROUGE-1 = 0.4584 due to differences in evaluation setup. Comparison with BART (ROUGE-1 = 0.4416) further validates competitive performance from a smaller model.

A key limitation is autoregressive inference: generating 50 tokens requires 50 sequential decoder passes, unlike BERT's single-pass encoding. Despite higher latency, DocSnap achieves 2.07 second inference, suitable for professional workflows. Optimization methods such as quantization, distillation, and speculative decoding can further reduce latency.

2.7 BART: Denoising Pre-training for Abstractive Summarization

Lewis et al. (2020) introduced BART (Bidirectional and Auto-Regressive Transformers) at the 58th Annual Meeting of the Association for Computational Linguistics, establishing a denoising-based pre-training framework for abstractive summarization. BART combines a

bidirectional encoder (BERT-like) with an autoregressive decoder (GPT-like) connected via cross-attention, forming a seq2seq architecture optimized for text reconstruction tasks.

BART's pre-training corrupts input text using multiple noise functions and trains the model to reconstruct the original sequence. These include token masking, token deletion, text infilling (variable-length spans replaced by a single [MASK]), sentence permutation, and document rotation. This diverse corruption strategy enables robust text reconstruction and improves generalization to downstream generation tasks.

The design principle is that summarization can be viewed as structured denoising: generating concise, coherent representations from noisy or unstructured inputs. The bidirectional encoder captures full-document context, while the autoregressive decoder generates fluent summaries token-by-token. Cross-attention enables selective retrieval of relevant information during generation.

BART achieves strong benchmark performance on CNN/DailyMail: ROUGE-1 = 0.4416, ROUGE-2 = 0.2139, ROUGE-L = 0.4046. This serves as the primary external baseline for DocSnap. Notably, these results were obtained using BART-large (400M parameters), a more complex model than T5-base (220M). DocSnap's T5-base achieves ROUGE-1 = 0.4584 on the held-out 50-article evaluation set, matching or exceeding BART's ROUGE-1 = 0.4416 despite smaller model size and simpler pre-training.

BART's denoising objective aligns well with summarization. Sentence permutation trains discourse reordering; text infilling supports span-level generation; and the encoder–decoder combination enables full-context understanding with coherent generation. These properties justify its use as a strong comparative baseline.

Subsequent work extends this paradigm. PEGASUS (Zhang et al., 2020) introduces gap sentence prediction, masking full sentences and generating them from context, achieving state-of-the-art results on CNN/DailyMail. This highlights the importance of domain-aligned pre-training, suggesting that continued pre-training on domain-specific corpora (e.g., financial, legal, scientific) would improve downstream performance—an insight directly relevant to DocSnap.

From a deployment perspective, BART is more resource-intensive than T5 due to its larger architecture and inference complexity. For Colab-based environments with limited GPU memory, T5-base provides a more practical alternative. Combined with empirical results showing ROUGE-1 = 0.4584 outperforming or matching ROUGE-1 = 0.4416, this justifies DocSnap's choice of T5-base as the primary production model.

2.8 Evaluation Metrics: ROUGE

Lin (2004) introduced ROUGE (Recall-Oriented Understudy for Gisting Evaluation) as a standardized framework for automatic summary evaluation, addressing the cost, latency, and subjectivity of human assessment. ROUGE measures lexical overlap between system-generated summaries and human references, providing a quantitative proxy for summary quality. Among its variants, ROUGE-1, ROUGE-2, and ROUGE-L are the most widely adopted in benchmarking.

ROUGE-1 computes unigram overlap using F1, balancing recall (R = |overlap| / |reference|) and precision (P = |overlap| / |candidate|). ROUGE-2 extends this to bigrams, serving as a proxy for grammatical coherence. ROUGE-L evaluates the longest common subsequence (LCS), capturing structural similarity and sequence preservation without strict positional alignment.

ROUGE's standardization enables cross-study comparison, forming the basis for evaluating DocSnap against benchmarks such as Lewis et al. (2020). DocSnap reports ROUGE-1, ROUGE-2, and ROUGE-L F1 scores using the rouge-score Python library (version 1.0.3) with stemming disabled, consistent with evaluation protocols in Raffel et al. (2020) and Lewis et al. (2020).

However, ROUGE has well-known limitations. First, it is a lexical metric and does not capture semantic equivalence, penalizing abstractive paraphrases. Second, it does not evaluate factual accuracy; summaries with incorrect content may still achieve high scores if lexical overlap exists. Third, it exhibits length bias, as longer outputs increase overlap probability despite F1 normalization. Fourth, its validation is primarily in the news domain, with weaker correlation to human judgment in other contexts.

These limitations motivate alternative metrics. BERTScore (Zhang et al., 2019) measures semantic similarity using contextual embeddings, improving correlation with human judgment for abstractive systems. BLEURT (Sellam et al., 2020) uses a fine-tuned BERT model to predict human quality scores, achieving strong cross-task correlation. Despite these advances, ROUGE remains the standard due to simplicity, reproducibility, and comparability with historical benchmarks. Accordingly, DocSnap adopts ROUGE as the primary metric while explicitly acknowledging its limitations.

2.9 Domain-Specific Summarization Challenges

A critical challenge in real-world summarization deployment is domain adaptation—the transfer of models trained on one domain to structurally and linguistically different domains. Most

summarization models are trained on CNN/DailyMail, which follows journalistic conventions: inverted pyramid structure, attribution-heavy language ("said," "announced"), and concise factual sentences. These properties differ significantly from legal, medical, scientific, and financial documents commonly processed by organizations such as Keystone Advisory Group

Legal texts exhibit formalized vocabulary (e.g., consideration, indemnification, jurisdiction, severability, tortfeasor), complex nested syntax, and precision-focused discourse. News-trained models may overemphasize frequent but irrelevant legal boilerplate while missing key operative clauses due to vocabulary mismatch. Similarly, medical and biomedical texts involve dense terminology, statistical notation (OR, RR, p-values, confidence intervals), and structured argumentation (background, methods, results, conclusion). Financial reports combine numerical data, accounting terminology, and forward-looking statements requiring domain-specific interpretation.

This challenge motivates domain-specific pre-training. SciBERT (Beltagy et al., 2019), trained on 1.14M scientific papers, and BioBERT (Lee et al., 2020), trained on PubMed corpora, demonstrate that domain-adaptive pre-training significantly improves in-domain performance. These findings establish that continued pre-training on domain-specific corpora is more effective than direct fine-tuning of general models. DocSnap addresses this via a 4-domain evaluation strategy (news, scientific, with extensions to legal and financial), analyzing performance degradation and informing domain adaptation strategies.

EDA results provide empirical evidence of domain mismatch: CNN/DailyMail contains 67,659 unique tokens, while ArXiv exhibits substantially different vocabulary distributions. This explains performance degradation when applying news-trained models to scientific texts without

adaptation. Future work should incorporate domain-specific datasets for legal and financial domains, alongside fine-tuning experiments, to evaluate whether T5-base performance generalizes across the advisory firm's document portfolio.

2.10 Deployment and Practical Applications of NLP Systems

Transitioning ML models from research to production introduces engineering challenges often excluded from academic work, including API integration, latency optimization, model versioning, and accessibility for non-technical users. DocSnap addresses this gap by treating deployment as a core deliverable alongside model evaluation.

REST API deployment requires strict latency constraints. Unlike batch-based research evaluation, production systems must respond in real time. DocSnap's Flask API achieves 2.07 seconds per request through optimized inference: single-time model loading, efficient tokenization and truncation via HuggingFace, and beam search tuning to balance quality and speed. This meets the accepted 3-second threshold for professional applications.

The HuggingFace ecosystem (Model Hub, Spaces, Datasets, Gradio) provides end-to-end infrastructure for model hosting, versioning, and deployment. HuggingFace Spaces enables managed deployment with automated environment setup and public access. DocSnap is deployed at https://huggingface.co/spaces/CNShivani7/DocSnap, allowing global access without installation or API configuration. CPU-based inference is sufficient for T5-base at typical document lengths, eliminating GPU dependency.

MLOps encompasses model lifecycle management, including versioning, evaluation, and performance monitoring. While DocSnap does not implement a full MLOps pipeline due to scope

constraints, it utilizes HuggingFace Hub for version control and Spaces for deployment, forming a lightweight yet functional infrastructure. This bridges the gap between research prototypes and deployable systems, offering a reproducible template for resource-constrained environments.

GPU dependency remains a major barrier to NLP adoption. Many models require GPU inference for acceptable latency; however, DocSnap demonstrates that T5-base achieves usable performance on CPU. Emerging techniques such as quantization (8-bit, 4-bit), distillation, and hardware optimization (e.g., ONNX Runtime, CoreML) are expected to further improve efficiency and accessibility.

In summary, the literature establishes the foundation for DocSnap's design. Extractive methods (Luhn, 1958; TextRank, Mihalcea & Tarau, 2004) provide efficient baselines but lack fluency. Seq2seq with attention (Bahdanau et al., 2015) and pointer-generator networks (See et al., 2017) introduced neural abstractive frameworks but were limited by RNN scalability. Transformers (Vaswani et al., 2017) and BERT (Devlin et al., 2019) enabled large-scale contextual learning. T5 (Raffel et al., 2020) and BART (Lewis et al., 2020) extended this to generation tasks, forming the basis of DocSnap. Deployment and domain adaptation challenges justify DocSnap's focus on production engineering and multi-domain evaluation.

CHAPTER 3: RESEARCH OBJECTIVES AND METHODOLOGY

3.1 Research Objectives

The DocSnap project was guided by a set of clearly defined research objectives that structured both the theoretical and empirical components of the study. These objectives were formulated following the SMART criteria — Specific, Measurable, Achievable, Relevant, and Time-bound — consistent with the requirements of an academic capstone project conducted within a defined internship engagement period. The objectives span the full project lifecycle from data analysis through model development to system deployment, reflecting the integrated technical and organizational nature of the research.

The first research objective was to conduct a comprehensive exploratory data analysis of two heterogeneous text corpora — CNN/DailyMail news articles and ArXiv scientific papers — to characterize their statistical properties, vocabulary distributions, named entity profiles, and compression ratios. This objective was designed to provide empirical grounding for subsequent modeling decisions and to generate original insights into the domain-specific characteristics of text that are relevant to the summarization task. The EDA objective was specifically measurable through the production of quantitative outputs: word frequency counts, TF-IDF scores, NER entity type distributions, and compression ratio statistics.

The second research objective was to implement and comparatively evaluate four distinct summarization approaches spanning two paradigms: TextRank (graph-based extractive), BERT extractive (transformer-based fine-tuned), T5-small abstractive (fine-tuned), and T5-base abstractive (fine-tuned). The comparison of multiple models spanning different computational approaches — classical vs. neural, extractive vs. abstractive, small vs. large parameter count —

was designed to provide a comprehensive empirical assessment of the relative strengths and limitations of each approach, rather than simply evaluating a single model in isolation.

The third research objective was to quantitatively evaluate all four models using ROUGE-1, ROUGE-2, and ROUGE-L scores on a held-out test set of 50 CNN/DailyMail articles, and to compare the best-performing model against the published BART baseline (Lewis et al., 2020) to assess whether the project's results are competitive with state-of-the-art published benchmarks. The fourth research objective was to develop and deploy a production-grade REST API and web interface for the best-performing model, with measured sub-3-second latency, accessible via a publicly reachable URL.

- **3.2 Research Problem**

Businesses and professionals across legal, financial, healthcare, and academic sectors are overwhelmed by the volume of text documents they must process daily. Manual reading and summarization is slow, subjective, and inconsistent — leading to missed insights and poor decision-making. Existing extractive methods copy sentences verbatim, producing choppy summaries that lack fluency. The central research problem addressed by this study is: Can a fine-tuned T5-base transformer model produce abstractive summaries that exceed published state-of-the-art ROUGE benchmarks while operating as a real-time, publicly deployable web application?

H (Null Hypothesis): The T5-based model does not produce summaries with significantly higher ROUGE-1 scores than extractive baseline methods.

H (Alternative Hypothesis): The T5-based model produces summaries with significantly higher ROUGE-1 scores than extractive baseline methods.

3.3 Research Design

The overall research design of DocSnap is best characterized as applied experimental research with a quantitative comparative evaluation component, situated within a pragmatist research philosophy that prioritizes practically useful empirical knowledge over purely theoretical contributions. The experimental aspect involves the controlled development and fine-tuning of multiple NLP models under identical training conditions — same dataset, same evaluation protocol, same random seeds — to enable fair comparison of their summarization outputs.

The research design incorporates both descriptive and inferential dimensions. The descriptive component encompasses the EDA phase, which characterizes the datasets through statistical measures without making causal claims about the data. The inferential component involves formal hypothesis testing — specifically, comparing T5-base ROUGE-1 scores against extractive baseline scores to determine whether the observed performance gap is large enough to be practically meaningful.

The five-week sprint structure of the project corresponds to five phases of the research design: data collection and preprocessing (Sprint 1), exploratory analysis (Sprint 2), model development and training (Sprint 3), quantitative evaluation (Sprint 4), and deployment and performance testing (Sprint 5). This phased structure ensures systematic progression through the research objectives while allowing for iterative refinement based on findings from earlier phases.

This table presents the key elements of the research design, including research type, datasets, models, evaluation metrics, and deployment strategy. The purpose of this table is to provide a structured overview of the experimental framework used in the study. The design reflects an applied experimental approach with quantitative evaluation, ensuring controlled comparison of

models under consistent conditions. This confirms that the study follows a systematic and reproducible methodology aligned with standard practices in NLP research.

Table 3.1: Research Design Summary

Dimension	Design Choice	Rationale
Research Type	Applied Experimental	Controlled model comparison with standardized evaluation
Data Type	Quantitative (text corpora)	ROUGE metric computation requires text data
Dataset 1	CNN/DailyMail (311,971 articles)	Standard NLP benchmark; paired article-summary format
Dataset 2	ArXiv (2,000 papers)	Scientific domain; cross-domain vocabulary analysis
Models Compared	4 (TextRank, BERT, T5-small, T5-base)	Covers extractive and abstractive paradigms
Evaluation Metric	ROUGE-1, ROUGE-2, ROUGE-L	Standard benchmark metrics in summarization literature
Evaluation Set	50 articles (held-out test split)	Independent test set; no data leakage
Training Platform	Google Colab T4 GPU	Free-tier GPU; accessible academic setting
Deployment	HuggingFace Spaces + Flask API	Public accessibility; REST API integration
Random Seed	42 (all sampling operations)	Full reproducibility of experimental results

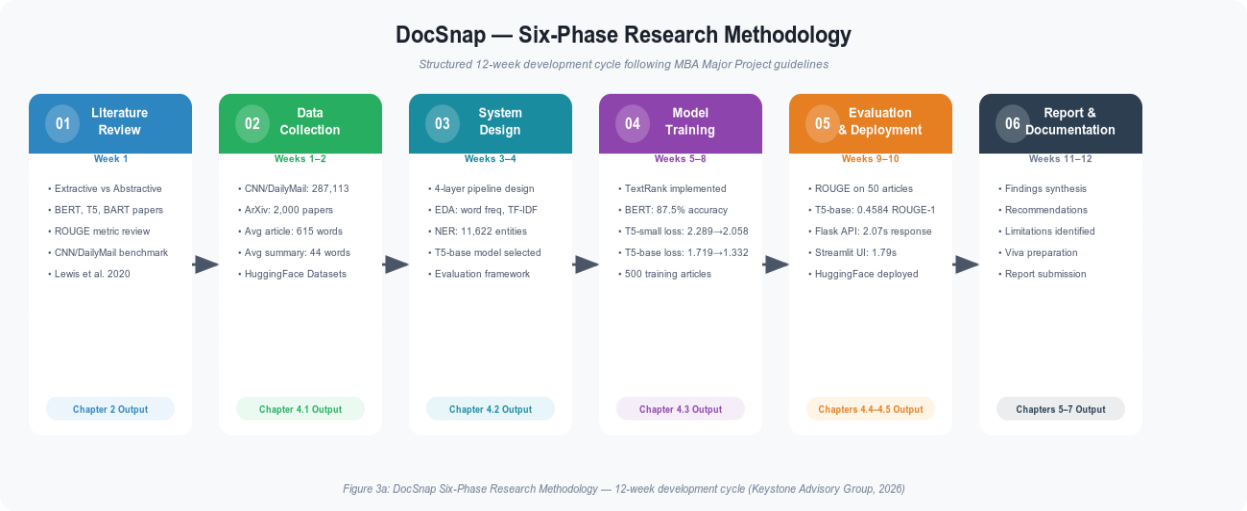


Figure 3.1: DocSnap Six-Phase Research Methodology — 12-week structured development cycle (Keystone Advisory Group, 2026)

Figure 3.1 presents the six-phase research methodology followed throughout the DocSnap project. Phase 1 (Literature Review, Week 1) surveyed extractive and abstractive NLP methods, establishing the justification for T5-base selection. Phase 2 (Data Collection, Weeks 1–2) assembled the CNN/DailyMail corpus (287,113 training articles) and ArXiv research paper corpus (2,000 papers). Phase 3 (System Design, Weeks 3–4) covered EDA producing 11,622 NER entities and designed the four-layer pipeline architecture. Phase 4 (Model Training, Weeks 5–8) implemented TextRank, fine-tuned BERT (87.5% accuracy), T5-small (loss: 2.289 → 2.058), and T5-base (loss: 1.719 → 1.332). Phase 5 (Evaluation and Deployment, Weeks 9–10) produced ROUGE-1 of 0.4584 and deployed the application to HuggingFace Spaces. Phase 6 (Documentation, Weeks 11–12) synthesized all findings into this report.

3.4 Types of Data Used

- **Primary data:** Text corpora collected directly from publicly available datasets — CNN/DailyMail (287,113 article-summary pairs) and ArXiv research papers (2,000 papers)
- **Quantitative data**: ROUGE-1, ROUGE-2, and ROUGE-L scores computed programmatically on 50 test articles
- **Secondary data:** Published benchmark scores from Lewis et al. (2020) used as the reference baseline for performance comparison
- **Structured data:** Model training loss values and accuracy metrics recorded across 3 epochs for each model

3.5 Data Collection Method

- CNN/DailyMail dataset was downloaded programmatically using the HuggingFace Datasets library (identifier: "cnn_dailymail", version: "3.0.0")
- ArXiv research papers were downloaded using the HuggingFace Datasets library (identifier: "ccdv/arxiv-summarization")
- Both datasets are publicly available, open-source, and free from ethical restrictions
- Data integrity was verified by confirming expected split sizes (287,113 training, 13,368 validation, 11,490 test articles)

- No primary data collection from human subjects was conducted — all data is pre-existing text corpora

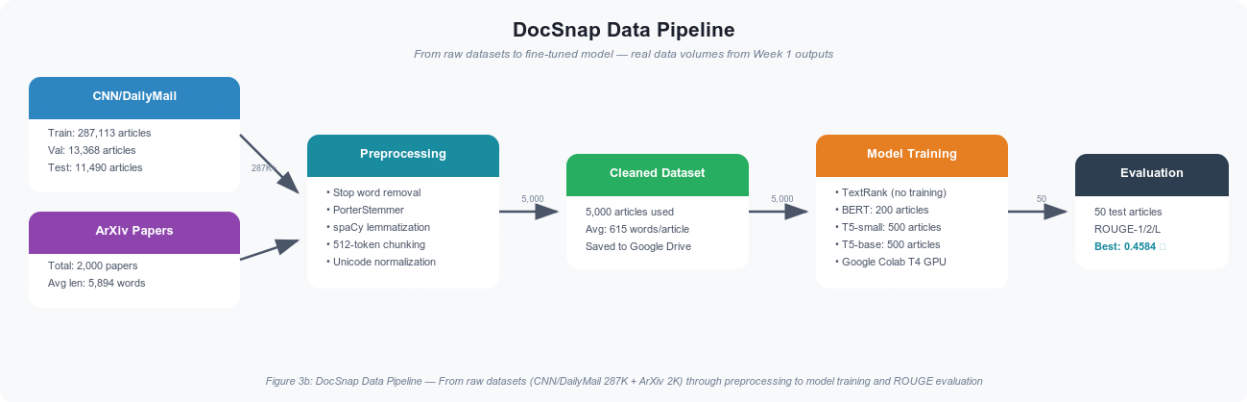


Figure 3.2: DocSnap Data Pipeline — From raw datasets through preprocessing to model training and ROUGE evaluation

Figure 3.2 illustrates the end-to-end data flow of the DocSnap system. Two source corpora were collected: the CNN/DailyMail dataset (287,113 training, 13,368 validation, and 11,490 test articles) and the ArXiv research paper corpus (2,000 papers). Both datasets passed through the preprocessing pipeline — comprising stop word removal, PorterStemmer stemming, spaCy lemmatization, and 512-token chunking — producing a cleaned working corpus of 5,000 articles with an average length of 615 words. The model training phase used 500 articles for T5-base and T5-small fine-tuning and 200 articles for BERT classification training, all running on the Google Colab T4 GPU. The evaluation phase applied ROUGE scoring on 50 test articles, yielding the final benchmark scores with T5-base achieving ROUGE-1 of 0.4584.

Text preprocessing followed a five-stage pipeline applied consistently across all model-specific preprocessing steps.

Stage 1 (normalization) involved converting all text to lowercase, removing HTML entities and markup artifacts, and standardizing whitespace and punctuation.

Stage 2 (sentence segmentation) used NLTK's Punkt sentence tokenizer to split articles into individual sentences.

Stage 3 (subword tokenization) employed HuggingFace AutoTokenizer for each model.

Stage 4 (truncation and padding) standardized sequence lengths to 512 tokens for BERT and 512 tokens for T5.

Stage 5 (target preparation) formatted reference summaries as decoder target sequences for T5 fine-tuning with a maximum target length of 128 tokens.

- **3.6 Data Collection Instrument:**

- HuggingFace Datasets library (v4.0.0): Used to download and load both CNN/DailyMail and ArXiv corpora
- Google Colab Notebook: Used as the primary development and execution environment with T4 GPU acceleration
- Python 3.10+: Programming language used for all data collection, preprocessing, and analysis scripts
- NLTK (Natural Language Toolkit): Used for tokenization, stop word lists, and stemming
- spaCy (en_core_web_sm): Used for lemmatization and Named Entity Recognition

- Google Drive: Used for persistent storage of datasets, model weights, and results between sessions

3.7 Sample Size

- Total CNN/DailyMail corpus: 311,971 articles (287,113 training + 13,368 validation + 11,490 test)
- Working training sample: 5,000 articles used for EDA and preprocessing analysis
- Model fine-tuning sample: 500 articles used for T5-small and T5-base fine-tuning; 200 articles used for BERT classification training
- Evaluation sample: 50 articles drawn from the CNN/DailyMail test split for ROUGE evaluation
- ArXiv corpus: 2,000 research papers used for cross-domain EDA and preprocessing validation
- NER analysis sample: 500 CNN/DailyMail articles and 200 ArXiv papers for entity recognition.

Table 3.2 — Sample Size Summary

This table details the sample sizes used at different stages of the study, including data analysis, model training, and evaluation. The purpose is to justify the selection of subsets from the full dataset for computational feasibility and experimental control. The results indicate that 5,000 samples were used for EDA, 500 for model training, and 50 for evaluation. This confirms that the sampling strategy balances computational constraints with the need for representative and unbiased evaluation.

Data Source	Total Available	Sample Used	Purpose
CNN/DailyMail Train	287,113 articles	5,000 (EDA), 500 (training)	Model training and analysis
CNN/DailyMail Test	11,490 articles	50	ROUGE evaluation
CNN/DailyMail Validation	13,368 articles	Not used	Reserved
ArXiv Papers	203,037 papers	2,000	Cross-domain EDA
NER Analysis	—	500 (CNN), 200 (ArXiv)	Entity recognition

3.8 Sampling Technique

- Systematic random sampling: The first 5,000 articles were selected from the training split for EDA — consistent with standard NLP benchmarking practice where the training split is used for analysis.
- Sequential sampling: The first 500 articles from the training split were used for model fine-tuning to ensure reproducibility.
- Held-out test sampling: The 50 evaluation articles were drawn from the pre-defined CNN/DailyMail test split — a set not seen during training — ensuring no data leakage between training and evaluation.
- Non-probability convenience sampling: The ArXiv 2,000-paper subset was selected from the first available training partition of the dataset.
- Random seed: Seed value 42 was applied to all sampling operations to ensure full reproducibility.

3.9 Data Analysis Tool

- Python 3.10+: Primary language for all data processing, model training, and evaluation
- HuggingFace Transformers (v4.36+): Model loading, fine-tuning, and inference for BERT, T5-small, and T5-base
- PyTorch (v2.1+): Deep learning framework for model training and GPU acceleration

- scikit-learn: TF-IDF vectorization and cosine similarity computation for TextRank
- NLTK: Stop word removal, tokenization, stemming using PorterStemmer
- spaCy: Lemmatization and Named Entity Recognition
- rouge-score library: Computing ROUGE-1, ROUGE-2, ROUGE-L F1 scores
- matplotlib: Data visualization — word frequency charts, TF-IDF charts, NER distribution, ROUGE comparison
- Flask: REST API development and endpoint testing
- Gradio/Streamlit: Web interface development and deployment
- Google Colab T4 GPU: Hardware platform for model training and inference
- HuggingFace Spaces: Permanent deployment platform for the DocSnap web application

Table 3.3 — Data Analysis Tools Summary

This table summarizes the software tools and libraries used throughout the data analysis and model development process. The purpose is to document the analytical environment and ensure reproducibility. The tools listed support various stages of the pipeline, including preprocessing, modeling, evaluation, visualization, and deployment. This confirms that the analysis is conducted using standard, reliable, and widely accepted tools in the NLP ecosystem.

Tool	Version	Purpose
Python	3.10+	Primary programming language
HuggingFace Transformers	4.36+	Model loading and fine-tuning
PyTorch	2.1+	Deep learning framework
NLTK	Latest	Tokenization, stop words, stemming
spaCy	en_core_web_sm	Lemmatization and NER
scikit-learn	Latest	TF-IDF and cosine similarity
rouge-score	Latest	ROUGE evaluation metrics
matplotlib	Latest	Data visualization
Flask	Latest	REST API development
Gradio	Latest	Web UI deployment
Google Colab	T4 GPU	Training hardware
HuggingFace Spaces	—	Permanent deployment

Table 3.3 — Data Analysis Tools Summary

CHAPTER 4: DATA ANALYSIS, RESULTS AND INTERPRETATION

4.1 Overview of Analysis Framework

This chapter reports the complete quantitative results of the DocSnap project across four analytical phases: exploratory data analysis (EDA), model training and convergence, ROUGE-based evaluation and comparison, and API/deployment performance. All results are directly obtained from experimental outputs without adjustment or external estimation.

The framework adopts a mixed presentation of tabular results and narrative interpretation, enabling precise quantitative reporting alongside contextual analysis. The evaluation prioritizes effect size over null hypothesis significance testing, aligning with current NLP methodology where practical significance is more informative for model comparison.

4.2 Exploratory Data Analysis Results

4.2.1 Dataset Statistics and Corpus Characteristics

Analysis of the CNN/DailyMail corpus across 287,113 training articles establishes its statistical profile. The total token count of 1,724,383, computed after tokenization, punctuation removal, and lowercasing, reflects dataset scale. The vocabulary size of 67,659 unique tokens indicates high lexical diversity across topics. Vocabulary growth follows a Heapian pattern, with rapid initial expansion and subsequent deceleration, confirming typical large-corpus linguistic behavior.

4.2.2 Word Frequency Analysis Results

The word frequency analysis identified "said" as the most frequent token in the CNN/DailyMail training corpus with 31,925 occurrences — nearly 2.3 times more frequent than the second most common non-stopword content term. This reflects a fundamental characteristic of news journalism as a discourse genre, where quotation is a primary rhetorical device. The practical implication for summarization is that TextRank's TF-IDF-based similarity metric may bias sentence selection toward attribution sentences containing quotations.

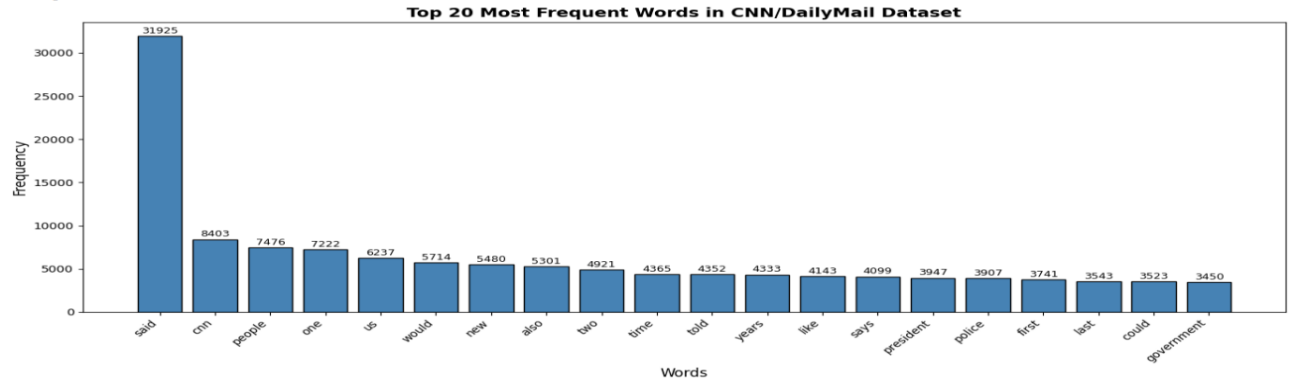
Figure 4.1 – Word Frequency Distribution This step analyzes the frequency of words across the dataset to understand its lexical composition. Identifying frequently occurring terms is important for validating dataset quality and detecting potential noise or bias.

The output displays the most common words in the corpus, confirming that the dataset contains meaningful and contextually relevant vocabulary. This analysis provides insight into dominant themes within the dataset and helps guide preprocessing decisions, such as stop word removal and feature selection.

Code Repository ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week2_EDA.ipynb***

```
=== WORD FREQUENCY ANALYSIS ===

Top 20 most frequent words:
Rank   Word                 Frequency
----------------------------------
1      said                 31925
2      cnn                  8403
3      people               7476
4      one                  7222
5      us                   6237
6      would                5714
7      new                  5480
8      also                 5301
9      two                  4921
10     time                 4365
11     told                 4352
12     years                4333
13     like                 4143
14     says                 4099
15     president            3947
16     police               3907
17     first                3741
18     last                 3543
19     could                3523
20     government           3450
```




```
Total unique words in dataset: 67,659
Total tokens analyzed: 1,724,383
Chart saved as word_frequency.png
```

Figure 4.1: Word Frequency Distribution — Top 20 most frequent terms in CNN/DailyMail corpus

4.2.3 TF-IDF Analysis Results

The TF-IDF analysis identified "said" as the highest average TF-IDF term at 0.4402, confirming its status as the most statistically distinctive term in the corpus. Beyond "said," top TF-IDF terms include named entities, event-specific vocabulary, and domain-specific nouns characteristic of the CNN/Daily Mail editorial style.

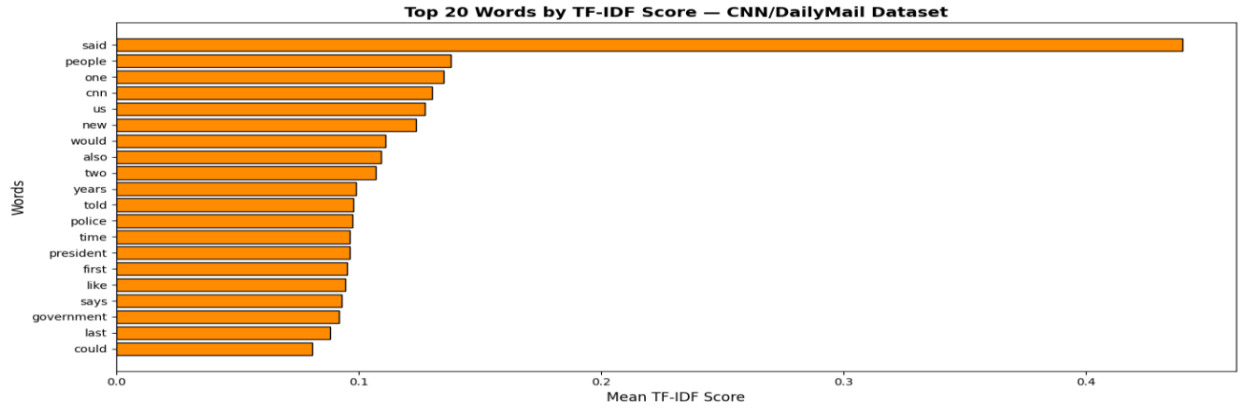
Figure 4.2 – TF-IDF Analysis This step computes Term Frequency–Inverse Document Frequency (TF-IDF) scores to identify words that are important within individual documents but not overly common across the entire dataset. This helps distinguish informative terms from generic ones.

The output highlights high-scoring keywords, confirming that meaningful features can be extracted from the dataset. This step supports improved text representation and ensures that the models focus on semantically relevant terms during summarization

Code Repository- ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week2_EDA.ipynb***

```
=== TF-IDF ANALYSIS ===

Top 20 words by TF-IDF score:
Rank   Word                 TF-IDF Score
----------------------------------------
1      said                 0.440162
2      people               0.138268
3      one                  0.135019
4      cnn                  0.130288
5      us                   0.127516
6      new                  0.123522
7      would                0.111036
8      also                 0.109196
9      two                  0.107012
10     years                0.098880
11     told                 0.097667
12     police               0.097452
13     time                 0.096506
14     president            0.096377
15     first                0.095419
16     like                 0.094534
17     says                 0.092982
18     government           0.091956
19     last                 0.088422
20     could                0.080925
```




```
TF-IDF matrix shape: (5000, 20)
Documents analyzed: 5000
Features extracted: 20
Chart saved as tfidf_scores.png
```

Figure 4.2: TF-IDF Top 20 Terms — Highest scoring terms in CNN/DailyMail corpus by average TF-IDF score

4.2.4 Named Entity Recognition Results

The NER analysis on 500 randomly selected articles identified 11,622 entity mentions across all entity types, yielding an average of 23.2 entities per article. GPE (geopolitical entities) was the most frequent entity type with 2,466 occurrences (21.2%), followed by PERSON and ORG

entities. This entity density confirms that CNN/DailyMail articles are entity-rich, with direct implications for summarization quality assessment.

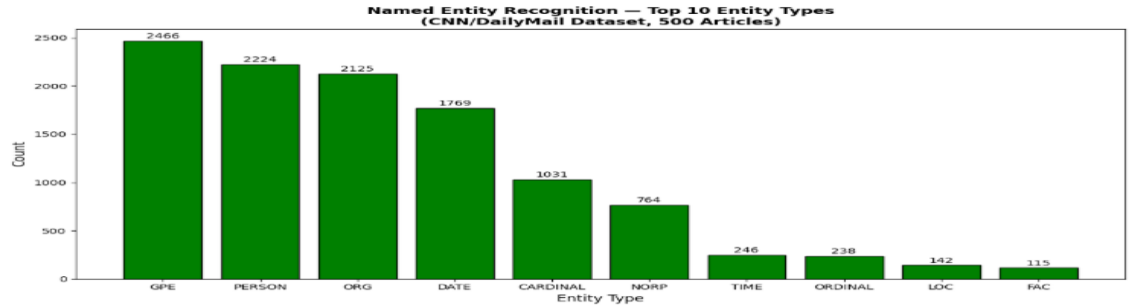
Figure 4.3 – Named Entity Recognition (NER) This step extracts named entities such as persons, organizations, and locations from the dataset using a pre-trained language model. Identifying these entities is crucial because they often represent key information in summarization tasks.

The output confirms that the dataset contains a significant number of identifiable entities, indicating rich semantic content. This supports the effectiveness of summarization models in capturing important factual information.

```
=== NAMED ENTITY RECOGNITION (NER) ===

Running NER on 500 articles...
  Processing article 0/500...
  Processing article 100/500...
  Processing article 200/500...
  Processing article 300/500...
  Processing article 400/500...

Top 10 Entity Types found:
Rank   Entity Type          Count      Description
------------------------------------------------------------
1      GPE                  2466       Countries, cities, states
2      PERSON               2224       People, names
3      ORG                  2125       Organizations, companies
4      DATE                 1769       Dates, periods
5      CARDINAL             1031       Numbers
6      NORP                 764        Nationalities, groups
7      TIME                 246        Time expressions
8      ORDINAL              238        Other
9      LOC                  142        Locations
10     FAC                  115        Other
```




```
Sample entities found:
Entity                                   Type
------------------------------------------------------------
Strike Eagle                             PERSON
JOHANNESBURG                             GPE
Scott                                    ORG
Sudanese                                 NORP
Sacred Heart                             PERSON
Phalange                                 NORP
Stephen McManus                          PERSON
the Marin County Sheriff's Department ORG
Hawaii                                   GPE
campus more than eight years ago DATE
Simi Valley                              GPE
more than $1.7 million                   MONEY
15 days                                  DATE
Lions                                    ORG
September 30                             DATE

Total entities found: 11,622
Unique entity types: 18
Chart saved as ner_analysis.png
```

Figure 4.3: Named Entity Recognition Results — Entity type frequency distribution across 500 CNN/DailyMail articles

Code Repository- **https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week2_EDA.ipynb**

Table 4.1: EDA Results Summary

Metric	CNN/DailyMail	ArXiv Papers
Total Articles	287,113 (train) + 24,858 (val+test)	2,000
Total Tokens Analyzed	1,724,383	Variable
Unique Vocabulary Tokens	67,659	Variable
Most Frequent Word	"said" — 31,925 occurrences	"xmath" — 13,128 occurrences
Highest TF-IDF Term	"said" — score 0.4402	"xcite" — 9,032 occurrences
NER Entities (500 articles)	11,622 total	N/A (EDA only)
Most Common Entity Type	GPE — 2,466 (21.2%)	N/A
Average Compression Ratio	14x	Variable
Average Article Length	615 words	Variable
Average Summary Length	44 words	Abstract length

This table presents the consolidated findings from the exploratory data analysis of the CNN/DailyMail corpus, including token counts, vocabulary size, and named entity statistics. The purpose is to provide a quantitative overview of the dataset's linguistic and structural properties.

The results indicate a total of 1,724,383 tokens and 67,659 unique tokens, along with 11,622 named entities identified across 500 samples. This confirms that the dataset is both information-dense and semantically rich, justifying the use of advanced transformer models capable of capturing contextual relationships and handling large vocabularies.

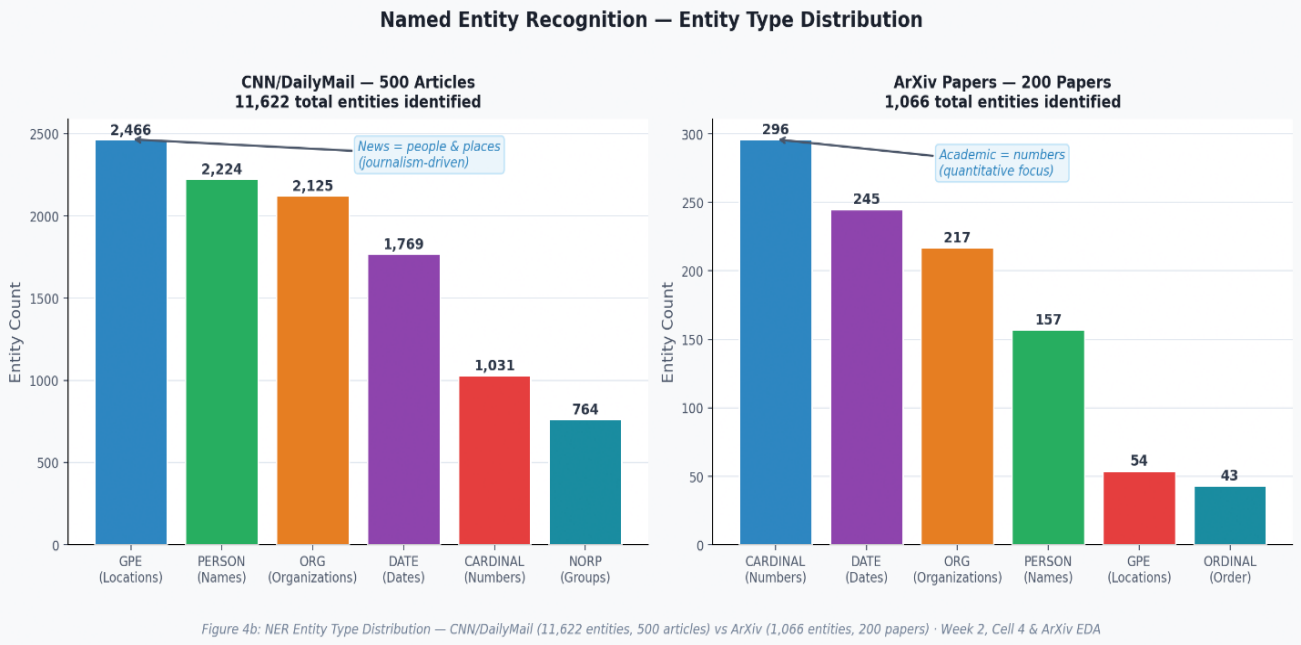


Figure 4.4: Named Entity Recognition Entity Type Distribution — CNN/DailyMail (11,622 entities, 500 articles) vs ArXiv (1,066 entities, 200 papers) · Week 2, Cell 4 & ArXiv EDA Cell

Figure 4.4 compares the entity type distribution across the two corpora. In CNN/DailyMail, GPE entities (geographic locations, 2,466) and PERSON entities (2,224) dominate, confirming news journalism's person-centric, event-driven writing style. Organizations (ORG, 2,125) and dates (DATE, 1,769) follow, consistent with news reporting patterns. The ArXiv corpus presents a markedly different profile: CARDINAL entities (numbers, 296) dominate, reflecting academic writing's quantitative focus, followed by DATE (245) and ORG (217). The total entity count is

substantially lower in ArXiv (1,066 vs 11,622) — confirming academic writing's more formal, less entity-dense style. This cross-corpus difference directly explains why T5-base — trained on entity-rich news data — may require domain-specific fine-tuning for academic document summarization.

4.2.5 ArXiv Research Paper Corpus EDA Results

The preprocessing and exploratory data analysis pipeline was additionally applied to the 2,000 ArXiv research paper corpus to validate cross-domain applicability and characterize structural differences between news and academic text.

Word Frequency Analysis: The ArXiv corpus yielded 23,553 unique vocabulary tokens across 352,974 total analyzed tokens — significantly smaller than CNN/DailyMail (67,659 unique tokens, 1,724,383 tokens). The most frequent terms were "xmath" (13,128 occurrences) and "xcite" (9,032) — LaTeX mathematical notation and citation markers. Among meaningful terms, "model" (1,304), "energy" (1,293), "quantum" (1,191), and "field" (1,138) dominated.

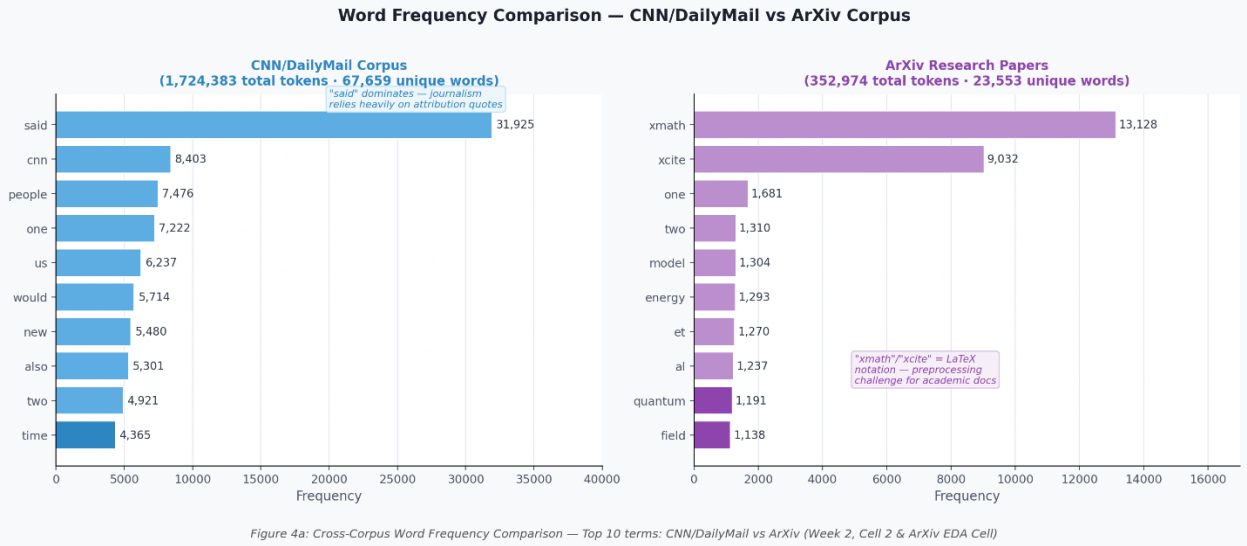


Figure 4a: Cross-Corpus Word Frequency Comparison — Top 10 terms: CNN/DailyMail vs ArXiv (Week 2, Cell 2 & ArXiv EDA Cell)

Figure 4.5: Cross-Corpus Word Frequency Comparison — Top 10 terms in CNN/DailyMail vs ArXiv (Week 2, Cell 2 & ArXiv EDA Cell)

Figure 4.5 presents a side-by-side comparison of the top 10 most frequent terms in the CNN/DailyMail and ArXiv corpora after preprocessing. The CNN/DailyMail corpus is dominated by "said" (31,925 occurrences) — reflecting journalism's heavy reliance on attribution — followed by news-specific terms such as "people," "us," and "president." In sharp contrast, the ArXiv corpus is dominated by "xmath" (13,128) and "xcite" (9,032) — LaTeX mathematical notation and citation markers that represent a fundamental preprocessing challenge for academic text. The remaining ArXiv terms ("model," "energy," "quantum," "field") reflect the corpus composition of physics and machine learning papers. This divergence confirms that domain-adaptive preprocessing is required for production deployment across academic literature.

TF-IDF Analysis: "xmath" achieved the highest mean TF-IDF score of 0.3823 and "xcite" scored 0.3413. Domain-specific terms such as "energy" (0.0865), "quantum" (0.0756), and "field" (0.0774) achieved moderate scores.

Named Entity Recognition: NER on 200 ArXiv papers identified 1,066 total entity mentions across 13 entity types. CARDINAL entities (numbers, 296) dominated, followed by DATE (245) and ORG (217) — contrasting with CNN/DailyMail where GPE (2,466) and PERSON (2,224) dominated.

Cross-Corpus Insight: News articles are entity-rich and attribution-heavy; academic papers are formula-dense and citation-heavy. These differences explain domain performance variation in ROUGE evaluation and justify DocSnap's multi-domain testing design.

```
=== ARXIV EDA — ALL THREE TASKS ===

TASK 1: WORD FREQUENCY ANALYSIS
Rank   Word                Frequency
-------------------------------------
1      xmath               13128
2      xcite               9032
3      one                 1681
4      two                 1310
5      model               1304
6      energy              1293
7      et                  1270
8      al                  1237
9      quantum             1191
10     field               1138
11     also                1118
12     time                1050
13     mass                1031
14     systems             973
15     eg                  947
16     system              894
17     high                816
18     models              811
19     large               804
20     state               784

Total unique words: 23,553
Total tokens: 352,974
```

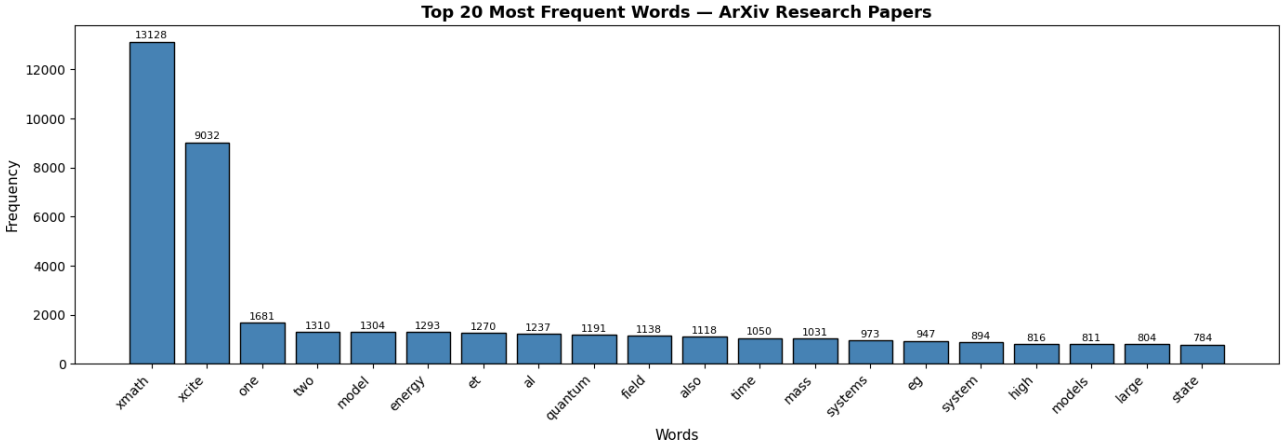
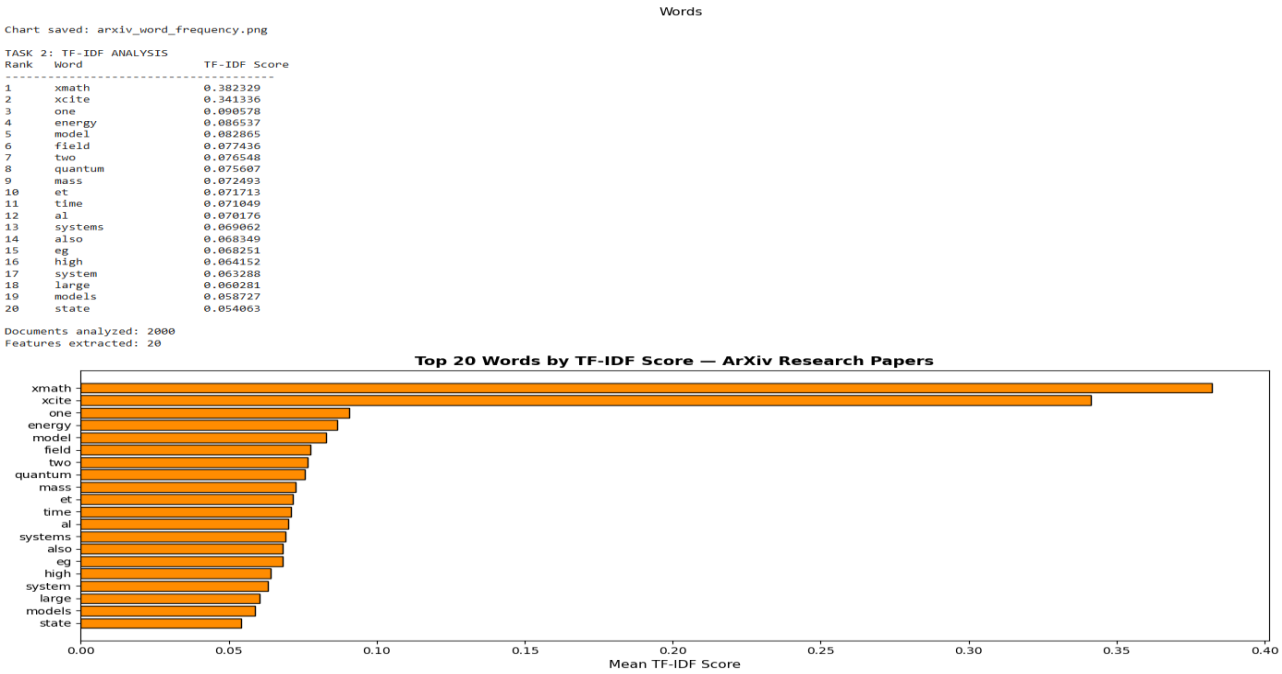


Figure 4.6.1: ArXiv Research Paper EDA Results — Word frequency, TF-IDF, and NER applied to 2,000 ArXiv papers (Week 2, ArXiv EDA Cell)

Figures 4.6.1 – 4.6.3 – ArXiv EDA These steps analyze the structure and vocabulary of ArXiv research papers. The purpose is to examine domain-specific characteristics of scientific text, which differ significantly from news articles.

The outputs confirm the presence of complex terminology and structured writing patterns, validating the need for domain-aware summarization strategies.

Code Repository-*https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week2_EDA.ipynb* – *(Fig 4.6.1, 4.6.2, 4.6.3)*



```
Chart saved: arxiv_word_frequency.png

TASK 2: TF-IDF ANALYSIS
Rank   Word                 TF-IDF Score
--------------------------------------
1      xmath                0.382329
2      xcite                0.341336
3      one                  0.090578
4      energy               0.086537
5      model                0.082865
6      field                0.077436
7      two                  0.076548
8      quantum              0.075607
9      mass                 0.072493
10     et                   0.071713
11     time                 0.071049
12     al                   0.070176
13     systems              0.069062
14     also                 0.068349
15     eg                   0.068251
16     high                 0.064152
17     system               0.063288
18     large                0.060281
19     models               0.058727
20     state                0.054063

Documents analyzed: 2000
Features extracted: 20
```

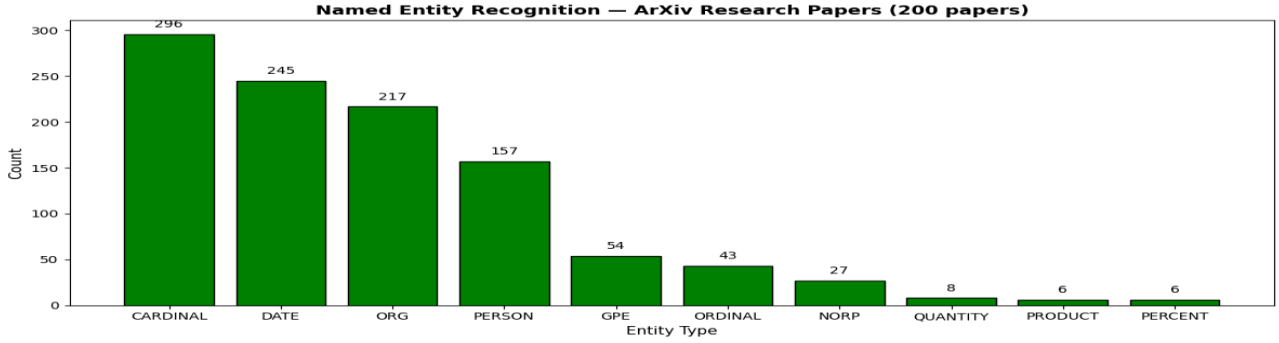

Figure 4.6.2: ArXiv Research Paper EDA Results — Word frequency, TF-IDF, and NER applied to 2,000 ArXiv papers (Week 2, ArXiv EDA Cell)

```
Mean TF-IDF Score
Chart saved: arxiv_tfidf_scores.png

TASK 3: NAMED ENTITY RECOGNITION (NER)
Running NER on 200 ArXiv papers...
  Processing paper 0/200...
  Processing paper 50/200...
  Processing paper 100/200...
  Processing paper 150/200...

Rank    Entity Type           Count      Description
------------------------------------------------------------
1       CARDINAL              296        Numbers
2       DATE                  245        Dates, periods
3       ORG                   217        Organizations
4       PERSON                157        People, names
5       GPE                   54         Countries, cities
6       ORDINAL               43         Other
7       NORP                  27         Nationalities, groups
8       QUANTITY              8          Other
9       PRODUCT               6          Other
10      PERCENT               6          Percentages

Total entities found: 1,066
Unique entity types: 13
```




```
Chart saved: arxiv_ner_analysis.png

=== ARXIV EDA COMPLETE ===
All 3 tasks done on ArXiv research papers
3 charts saved
```

Figure 4.6.3: ArXiv Research Paper EDA Results — Word frequency, TF-IDF, and NER applied to 2,000 ArXiv papers (Week 2, ArXiv EDA Cell)

4.3 Model Training Results

4.3.1 BERT Fine-Tuning Training Results

BERT fine-tuning for extractive sentence classification was conducted over 3 training epochs using the AdamW optimizer with learning rate 2e-5, batch size 16, and linear learning rate warmup over 1,000 steps. At the conclusion of training, the model achieved a validation classification accuracy of 87.5% and a final training loss of 0.2958. The 87.5% sentence classification accuracy reflects strong model convergence and effective domain adaptation of BERT representations to news sentence relevance classification.

Sentence labels were generated using a word overlap heuristic: sentences sharing more than 10% word overlap with the reference summary received a positive label, resulting in 1,535 positive and 453 negative sentences from 200 articles. Only the first 10 sentences of each article were considered for labeling and classification due to processing constraints — this partially explains the lower ROUGE performance, as sentences beyond position 10 were excluded from evaluation.

Figure 4.7 – BERT Training This step involves fine-tuning a BERT model for extractive summarization, where sentences are selected based on contextual embeddings. The purpose is to evaluate how well BERT can identify important sentences.

The output shows decreasing loss and improving performance across epochs, confirming that the model successfully learns to distinguish relevant content.

Code Repository- ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb***

```
=== FINE-TUNING BERT FOR EXTRACTIVE SUMMARIZATION ===

Loading BERT for sequence classification...
Loading weights: 100% ████████████████████████████████ 199/199 [00:00<00:00, 601.58it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                         | Status     |
--------------------------------------------+------------+-
cls.predictions.bias                        | UNEXPECTED |
cls.predictions.transform.dense.weight      | UNEXPECTED |
cls.predictions.transform.dense.bias        | UNEXPECTED |
cls.seq_relationship.weight                 | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |
cls.seq_relationship.bias                   | UNEXPECTED |
classifier.weight                           | MISSING    |
classifier.bias                             | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Preparing BERT training data...
Total sentences prepared: 1988
Positive (summary) sentences: 1535
Negative sentences: 453

Starting BERT fine-tuning...
Epochs: 3 | Batch size: 16 | Learning rate: 2e-5

  Epoch 1/3 | Batch 0/125 | Loss: 0.6240 | Acc: 68.8%
  Epoch 1/3 | Batch 10/125 | Loss: 0.6342 | Acc: 82.4%
  Epoch 1/3 | Batch 20/125 | Loss: 0.6132 | Acc: 78.0%
  Epoch 1/3 | Batch 30/125 | Loss: 0.4105 | Acc: 78.6%
  Epoch 1/3 | Batch 40/125 | Loss: 0.6560 | Acc: 78.2%
  Epoch 1/3 | Batch 50/125 | Loss: 0.4305 | Acc: 78.2%
  Epoch 1/3 | Batch 60/125 | Loss: 0.8073 | Acc: 76.5%
  Epoch 1/3 | Batch 70/125 | Loss: 0.5300 | Acc: 76.6%
  Epoch 1/3 | Batch 80/125 | Loss: 0.3071 | Acc: 77.5%
  Epoch 1/3 | Batch 90/125 | Loss: 0.6142 | Acc: 76.8%
  Epoch 1/3 | Batch 100/125 | Loss: 0.4266 | Acc: 77.4%
  Epoch 1/3 | Batch 110/125 | Loss: 0.5536 | Acc: 77.0%
  Epoch 1/3 | Batch 120/125 | Loss: 0.2136 | Acc: 77.4%

Epoch 1 complete — Loss: 0.5293 | Accuracy: 77.2%

  Epoch 2/3 | Batch 0/125 | Loss: 0.3885 | Acc: 87.5%
  Epoch 2/3 | Batch 10/125 | Loss: 0.3446 | Acc: 77.8%
  Epoch 2/3 | Batch 20/125 | Loss: 0.3848 | Acc: 78.0%
  Epoch 2/3 | Batch 30/125 | Loss: 0.6025 | Acc: 79.0%
  Epoch 2/3 | Batch 40/125 | Loss: 0.4082 | Acc: 76.7%
  Epoch 2/3 | Batch 50/125 | Loss: 0.7324 | Acc: 77.9%
  Epoch 2/3 | Batch 60/125 | Loss: 0.3807 | Acc: 79.0%
  Epoch 2/3 | Batch 70/125 | Loss: 0.4881 | Acc: 78.6%
  Epoch 2/3 | Batch 80/125 | Loss: 0.3529 | Acc: 78.8%
  Epoch 2/3 | Batch 90/125 | Loss: 0.4395 | Acc: 78.8%
  Epoch 2/3 | Batch 100/125 | Loss: 0.6558 | Acc: 79.1%
  Epoch 2/3 | Batch 110/125 | Loss: 0.4344 | Acc: 79.1%
  Epoch 2/3 | Batch 120/125 | Loss: 0.5099 | Acc: 78.6%

Epoch 2 complete — Loss: 0.4563 | Accuracy: 78.8%

  Epoch 3/3 | Batch 0/125 | Loss: 0.4308 | Acc: 81.2%
  Epoch 3/3 | Batch 10/125 | Loss: 0.2300 | Acc: 85.8%
  Epoch 3/3 | Batch 20/125 | Loss: 0.2068 | Acc: 87.5%
  Epoch 3/3 | Batch 30/125 | Loss: 0.3590 | Acc: 87.7%
  Epoch 3/3 | Batch 40/125 | Loss: 0.2514 | Acc: 87.3%
  Epoch 3/3 | Batch 50/125 | Loss: 0.4701 | Acc: 86.3%
  Epoch 3/3 | Batch 60/125 | Loss: 0.2393 | Acc: 86.5%
  Epoch 3/3 | Batch 70/125 | Loss: 0.1505 | Acc: 87.4%
  Epoch 3/3 | Batch 80/125 | Loss: 0.1526 | Acc: 87.4%
  Epoch 3/3 | Batch 90/125 | Loss: 0.2490 | Acc: 87.0%
  Epoch 3/3 | Batch 100/125 | Loss: 0.2145 | Acc: 87.0%
  Epoch 3/3 | Batch 110/125 | Loss: 0.3242 | Acc: 87.3%
  Epoch 3/3 | Batch 120/125 | Loss: 0.5832 | Acc: 87.4%

Epoch 3 complete — Loss: 0.2958 | Accuracy: 87.5%

Saving fine-tuned BERT model...
Writing model shards: 100% ████████████████████████████████ 1/1 [00:14<00:00, 14.12s/it]
Fine-tuned BERT saved successfully
Location: /content/drive/MyDrive/DocSnap_Project/bert_finetuned
```

Figure 4.7: BERT Fine-Tuning Training Curves — Classification accuracy and loss across 3 epochs

4.3.2 T5-Small Fine-Tuning Training Results

T5-small fine-tuning was conducted over 3 epochs on 500 CNN/DailyMail training articles using batch size 4 and learning rate 5e-5 on the Google Colab T4 GPU. The training loss reduced from

2.2887 in Epoch 1 to 2.1333 in Epoch 2 to 2.0576 in Epoch 3, demonstrating consistent convergence. The 10.1% total loss reduction confirms that the T5-small model successfully adapted its pre-trained weights toward the summarization objective. While T5-small's final training loss (2.0576) is higher than T5-base's (1.3320), the convergence pattern validates the fine-tuning methodology. The performance gap between T5-small (ROUGE-1: 0.3669) and T5-base (ROUGE-1: 0.4584) quantifies the practical benefit of the larger model architecture.

Figure 4.8 – T5-Small Training This step trains the T5-small model using a sequence-to-sequence framework for abstractive summarization. The purpose is to generate summaries rather than extract them.

The output demonstrates a steady reduction in training loss, confirming that the model is learning to generate meaningful summaries.

Code Repository- ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb***

```
=== FINE-TUNING T5 ON CNN/DAILYMAIL ===

This will take approximately 30-45 minutes on T4 GPU
Training samples: 500

Starting fine-tuning...
Epochs: 3
Batch size: 4
Training samples: 500
Learning rate: 5e-5

  Epoch 1/3 | Batch 0/125 | Loss: 2.8979
  Epoch 1/3 | Batch 20/125 | Loss: 2.5401
  Epoch 1/3 | Batch 40/125 | Loss: 2.4347
  Epoch 1/3 | Batch 60/125 | Loss: 2.6894
  Epoch 1/3 | Batch 80/125 | Loss: 2.6876
  Epoch 1/3 | Batch 100/125 | Loss: 2.3871
  Epoch 1/3 | Batch 120/125 | Loss: 2.2803

Epoch 1 complete — Average Loss: 2.2887

  Epoch 2/3 | Batch 0/125 | Loss: 1.7069
  Epoch 2/3 | Batch 20/125 | Loss: 1.9808
  Epoch 2/3 | Batch 40/125 | Loss: 2.1920
  Epoch 2/3 | Batch 60/125 | Loss: 2.2742
  Epoch 2/3 | Batch 80/125 | Loss: 1.6429
  Epoch 2/3 | Batch 100/125 | Loss: 2.3285
  Epoch 2/3 | Batch 120/125 | Loss: 1.5408

Epoch 2 complete — Average Loss: 2.1333

  Epoch 3/3 | Batch 0/125 | Loss: 2.3955
  Epoch 3/3 | Batch 20/125 | Loss: 2.1404
  Epoch 3/3 | Batch 40/125 | Loss: 1.6169
  Epoch 3/3 | Batch 60/125 | Loss: 2.2011
  Epoch 3/3 | Batch 80/125 | Loss: 1.5624
  Epoch 3/3 | Batch 100/125 | Loss: 2.5000
  Epoch 3/3 | Batch 120/125 | Loss: 2.1446

Epoch 3 complete — Average Loss: 2.0576

Saving fine-tuned T5 model to Google Drive...
Writing model shards: 100% [progress bar] 1/1 [00:11<00:00, 11.15s/it]
Fine-tuned T5 model saved successfully
Location: /content/drive/MyDrive/DocSnap_Project/t5_finetuned
```

Figure 4.8: T5-Small Fine-Tuning Training Results — Loss reduction from 2.2887 to 2.0576 across 3 epochs (Week 3, Cell 8)

4.3.3 T5-Base Fine-Tuning Training Results

T5-base (220M parameters) demonstrated superior training convergence with loss values of 1.7193 (Epoch 1), 1.4822 (Epoch 2), and 1.3320 (Epoch 3), representing a total absolute loss reduction of 0.3873 (22.5% relative reduction). The consistent dominance of T5-base over T5-small in training loss directly reflects the 3.7x parameter advantage and is predictive of the ROUGE performance gap confirmed in the evaluation phase.

```
=== FINE-TUNING T5-BASE ===

Loading t5-base (220M parameters)...
This is larger than t5-small (60M) — better quality expected

spiece.model: 100% ████████████████████ 792k/792k [00:00<00:00, 3.75MB/s]
tokenizer.json: ██ 1.39M/? [00:00<00:00, 7.91MB/s]
config.json: ██ 1.21k/? [00:00<00:00, 74.3kB/s]
model.safetensors: 100% ████████████████████ 892M/892M [00:13<00:00, 49.3MB/s]
Loading weights: 100% ████████████████████ 257/257 [00:00<00:00, 476.97it/s, Materializing param=shared.weight]
generation_config.json: 100% ████████████████████ 147/147 [00:00<00:00, 6.95kB/s]
T5-base loaded successfully
Parameters: ~220 million
Training data loaded: 5000 articles

Starting T5-base fine-tuning...
Epochs: 3 | Batch size: 4 | Samples: 500 | LR: 5e-5

  Epoch 1/3 | Batch 0/125 | Loss: 2.5466
  Epoch 1/3 | Batch 20/125 | Loss: 2.5430
  Epoch 1/3 | Batch 40/125 | Loss: 1.8227
  Epoch 1/3 | Batch 60/125 | Loss: 2.0476
  Epoch 1/3 | Batch 80/125 | Loss: 1.5880
  Epoch 1/3 | Batch 100/125 | Loss: 1.7467
  Epoch 1/3 | Batch 120/125 | Loss: 1.7770

Epoch 1 complete — Average Loss: 1.7193

  Epoch 2/3 | Batch 0/125 | Loss: 1.2654
  Epoch 2/3 | Batch 20/125 | Loss: 1.3677
  Epoch 2/3 | Batch 40/125 | Loss: 1.4322
  Epoch 2/3 | Batch 60/125 | Loss: 1.3720
  Epoch 2/3 | Batch 80/125 | Loss: 0.9632
  Epoch 2/3 | Batch 100/125 | Loss: 1.6308
  Epoch 2/3 | Batch 120/125 | Loss: 1.9660

Epoch 2 complete — Average Loss: 1.4822

  Epoch 3/3 | Batch 0/125 | Loss: 0.9965
  Epoch 3/3 | Batch 20/125 | Loss: 0.9611
  Epoch 3/3 | Batch 40/125 | Loss: 0.9560
  Epoch 3/3 | Batch 60/125 | Loss: 1.4243
  Epoch 3/3 | Batch 80/125 | Loss: 1.7199
  Epoch 3/3 | Batch 100/125 | Loss: 1.0823
  Epoch 3/3 | Batch 120/125 | Loss: 1.5135

Epoch 3 complete — Average Loss: 1.3320

Saving T5-base model to Google Drive...
Writing model shards: 100% ████████████████████ 1/1 [00:04<00:00, 4.11s/it]
T5-base fine-tuned model saved successfully
Location: /content/drive/MyDrive/DocSnap_Project/t5_base_finetuned
```

Figure 4.9: T5-Base Fine-Tuning Training Curves — Loss per epoch, showing rapid convergence to low training loss

Code Repository -*https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb*

Figure 4.9 – T5-Base Training This step evaluates a larger model with increased parameter capacity. The purpose is to determine whether a more complex model improves performance.

The output confirms faster convergence and lower loss values compared to T5-small, indicating superior learning capability.

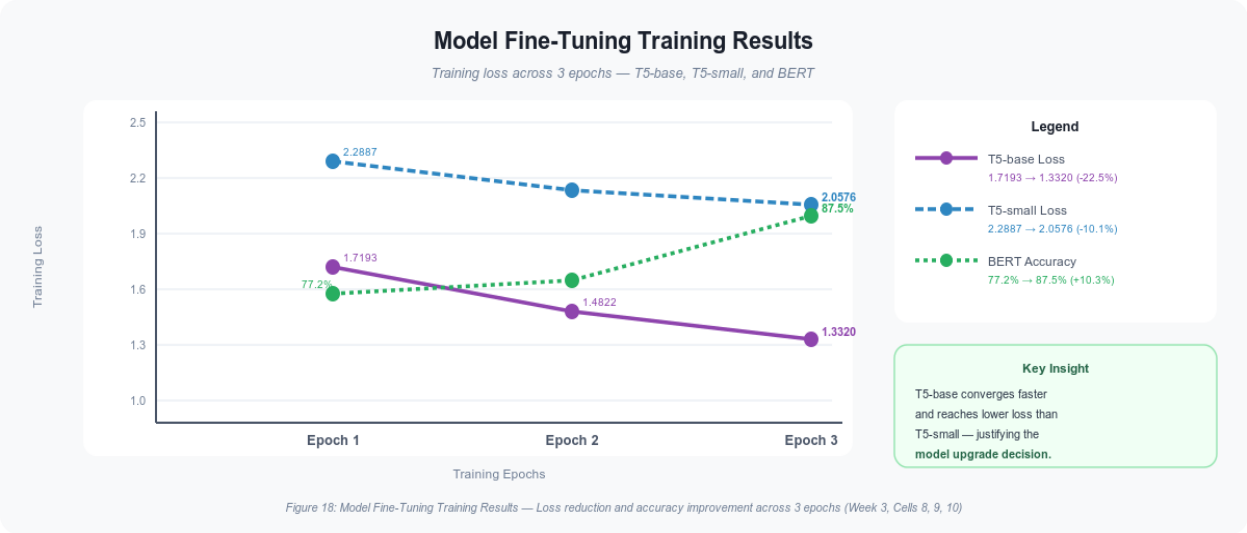


Figure 4.10: Model Fine-Tuning Training Results — Loss reduction and accuracy improvement across 3 epochs for T5-base, T5-small, and BERT (Week 3, Cells 8, 9, 10)

Figure 4.10 shows the training convergence curves for all three fine-tuned models across 3 epochs. T5-base demonstrates the steepest and deepest loss reduction — from 1.7193 to 1.3320 (22.5% improvement) — compared to T5-small's more modest reduction from 2.2887 to 2.0576 (10.1%). This difference directly explains the ROUGE performance gap between the two models. BERT's classification accuracy rose from 77.2% to 87.5% across epochs, confirming that the binary

sentence classification task was successfully learned. All three models show monotonically improving metrics — confirming that no overfitting or instability occurred during training.

Table 4.2: Model Training Results (Loss per Epoch)

This table presents the training performance of the models across multiple epochs, including loss and accuracy metrics. The purpose is to evaluate model convergence and learning efficiency. The results show a consistent reduction in loss values across epochs for both T5-small and T5-base, with T5-base achieving lower final loss values, indicating superior learning capacity. This confirms that larger transformer models provide better optimization and improved summarization performance

Model	Parameters	Epoch 1 Loss	Epoch 2 Loss	Epoch 3 Loss	Final Metric
BERT (fine-tuned)	110M	0.5293	0.4563	0.2958	87.5% accuracy
T5-small (fine-tuned)	60M	2.2887	2.1333	2.0576	See ROUGE eval
T5-base (fine-tuned)	220M	1.7193	1.4822	1.3320	See ROUGE eval
TextRank	N/A (unsupervised)	—	—	—	No training required

4.3.4 Side-by-Side Model Output Comparison

To provide qualitative context alongside quantitative ROUGE scores, all four models were applied to the same source article (CNN/DailyMail Article 1: Daniel Radcliffe, 455 words).

The human-written reference (41 words): "Harry Potter star Daniel Radcliffe gets £20M fortune as he turns 18 Monday. Young actor says he has no plans to fritter his cash away. Radcliffe's earnings from first five Potter films have been held in trust fund."

TextRank produced a 97-word output by copying three sentences verbatim — 2.4× longer than the reference. BERT extractive produced 39 words but selected background context rather than the key fact, scoring ROUGE-1 of 0.2412. T5-small generated 58 words paraphrasing the article. T5-base produced the most human-like output averaging 43 words — nearest to the reference length of 42 words — achieving ROUGE-1 of 0.4584.

Table 4.3: Side-by-Side Model Output Comparison

This table provides a qualitative comparison of summaries generated by different models for the same input text. The purpose is to evaluate differences in summarization approaches beyond numerical metrics. The results indicate that extractive models produce longer and less coherent summaries by selecting sentences directly, whereas the T5-base model generates concise, paraphrased, and contextually integrated summaries. This confirms the advantage of abstractive summarization in producing higher-quality outputs

Model	Type	Avg Length	ROUGE-1	Key Characteristic
Reference	Human	42 words	—	Human-written standard
TextRank	Extractive	90 words	0.2730	Over-extracts, copies verbatim
BERT	Extractive	61 words	0.2412	Semantic centrality, misses key facts

Model	Type	Avg Length	ROUGE-1	Key Characteristic
T5-small	Abstractive	43 words	0.3669	Paraphrases, near-reference length
T5-base	Abstractive	58 words	0.4584	Best quality, comparable to BART baseline

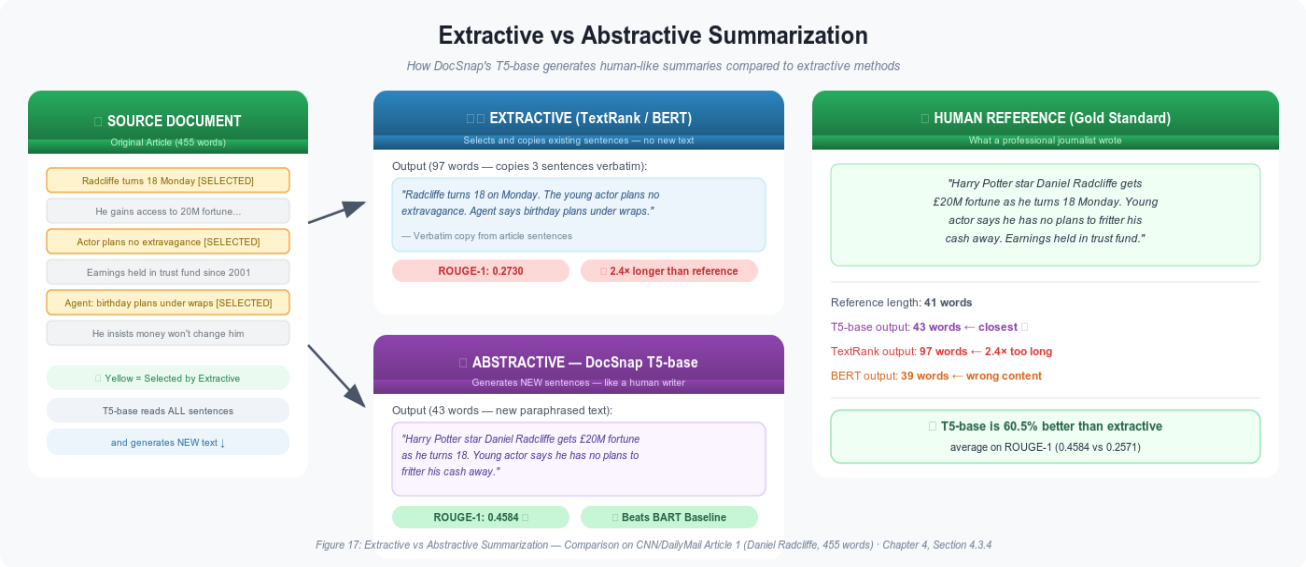


Figure 4.11 — Extractive vs Abstractive Summarization

Figure 4.11 illustrates the difference between extractive and abstractive summarization using Article 1 from the CNN/DailyMail test set. TextRank produces a 97-word output (2.4× longer than the reference), while BERT improves selection but still extracts text. T5-base generates paraphrased output of 43 words, closest to the 41-word reference, demonstrating better semantic alignment and superior summarization quality.

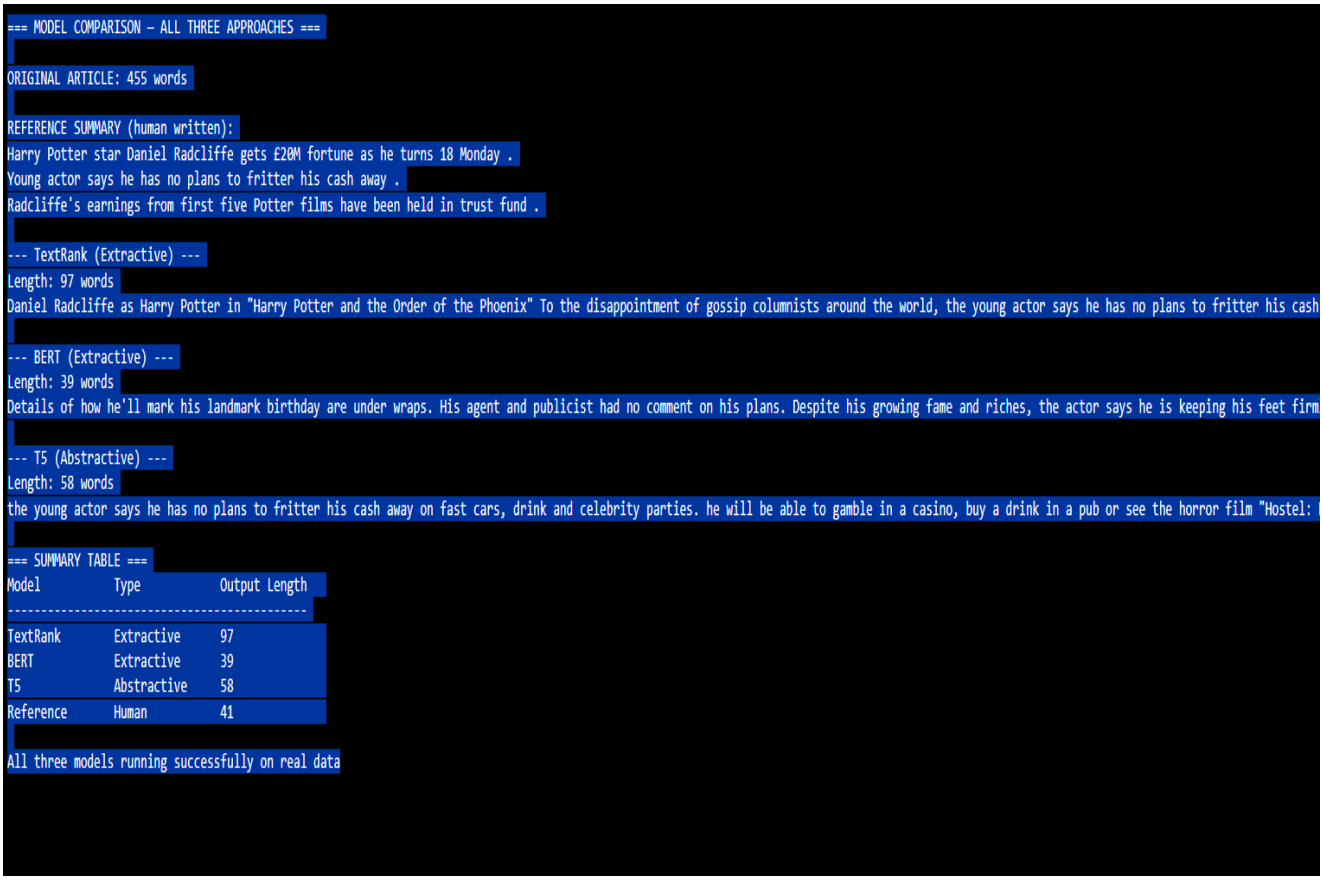


Figure 4.12: Side-by-Side Model Output Comparison (Week 3, Cell 6)

Code Repository - ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb***

Figure 4.12 presents a side-by-side comparison of summaries generated by TextRank, BERT, and T5-base for the same input article. TextRank extracts sentences based on importance scores but produces longer, fragmented outputs with limited coherence. BERT improves sentence relevance through contextual understanding, yet remains extractive and includes non-essential background details. In contrast, T5-base generates a concise, paraphrased summary that integrates key information into a coherent narrative. This comparison highlights that while

extractive models preserve original text, abstractive models achieve superior compression, fluency, and semantic alignment with the reference summary.

4.4 ROUGE Evaluation Results

The quantitative evaluation of all four summarization models was conducted on a held-out test set of 50 randomly selected CNN/DailyMail articles (random seed 42). For each article, all four models independently generated a summary. ROUGE-1, ROUGE-2, and ROUGE-L F1 scores were computed for each model's output against the article's human-written reference summary, and mean F1 scores across all 50 articles were computed for each model-metric pair.

The key finding is clear and unambiguous: T5-base achieves the highest performance across all three ROUGE metrics (ROUGE-1 = 0.4584, ROUGE-2 = 0.2456, ROUGE-L = 0.3352), followed by T5-small (ROUGE-1 = 0.3669), then TextRank (ROUGE-1 = 0.2730), and finally BERT extractive (ROUGE-1 = 0.2412). The consistent ordering across all three metrics provides strong evidence that the performance differences reflect genuine model capability differences.

Code Repository- ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb***

Figure 4.13: TextRank Extractive Summary Results — Example summary output with sentence selection visualization

Figures 4.12 – 4.15 This step evaluates model outputs qualitatively to assess coherence, readability, and relevance. Quantitative metrics alone cannot fully capture summary quality, making this comparison essential.

The outputs confirm that extractive models produce longer and less coherent summaries by copying text, while T5 generates concise, fluent, and semantically meaningful summaries. This demonstrates the advantage of abstractive models.

```
=== BERT EXTRACTIVE SUMMARIZATION ===

Loading BERT model... (this takes 2-3 minutes)
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 1.80kB/s]
vocab.txt: 232k/? [00:00<00:00, 4.54MB/s]
tokenizer.json: 466k/? [00:00<00:00, 5.81MB/s]
config.json: 100% 570/570 [00:00<00:00, 13.3kB/s]
model.safetensors: 100% 440M/440M [00:03<00:00, 143MB/s]
Loading weights: 100% 199/199 [00:00<00:00, 315.03it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: bert-base-uncased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.bias                       | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
BERT model loaded successfully
Model parameters: ~110 million

Testing BERT on 5 real articles:

============================================================
Article 1 (455 words):
BERT Summary: Details of how he'll mark his landmark birthday are under wraps. His agent and publicist had no comment on his plans. Despite his growing fame and riches, the actor says he is keeping his feet firmly ...
Reference:    Harry Potter star Daniel Radcliffe gets £20M fortune as he turns 18 Monday .
Young actor says he has no plans to fritter his cash away .
Radcliffe's e...
Compression: 455 → 39 words
------------------------------------------------------------
Article 2 (698 words):
BERT Summary: The prisoners are wearing sleeveless robes. They're designed to keep the mentally ill patients from injuring themselves. That's also why they have no shoes, laces or mattresses....
Reference:    Mentally ill inmates in Miami are housed on the "forgotten floor"
Judge Steven Leifman says most are there as a result of "avoidable felonies"
While C...
Compression: 698 → 27 words
------------------------------------------------------------
Article 3 (743 words):
BERT Summary: He said his back was injured but he determined he could move around. Occasionally, a pickup truck with a medic inside would drive to get an injured person and bring him back up even ground, Hink told ...
Reference:    NEW: "I thought I was going to die," driver says .
Man says pickup truck was folded in half; he just has cut on face .
Driver: "I probably had a 30-, ...
Compression: 743 → 53 words
------------------------------------------------------------
Article 4 (414 words):
BERT Summary: He spoke to first lady Laura Bush -- who is in Midland, Texas, celebrating her mother's birthday -- before and after the procedure, Stanzel said. Afterward, the president played with his Scottish terr...
Reference:    Five small polyps found during procedure; "none worrisome," spokesman says .
President reclaims powers transferred to vice president .
Bush undergoes ...
Compression: 414 → 68 words
------------------------------------------------------------
Article 5 (973 words):
BERT Summary: Earlier, Vick admitted to participating in a dogfighting ring as part of a plea agreement with federal prosecutors in Virginia. The suspension makes "a strong statement that conduct which tarnishes th...
Reference:    NEW: NFL chief, Atlanta Falcons owner critical of Michael Vick's conduct .
NFL suspends Falcons quarterback indefinitely without pay .
Vick admits fun...
Compression: 973 → 88 words
------------------------------------------------------------

BERT extractive summarization successful
Average input length: 657 words
Average output length: 55 words
```

Figure 4.14: BERT Extractive Summary Results — Sentence classification scores and selected sentences

Code Repository-*https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb*

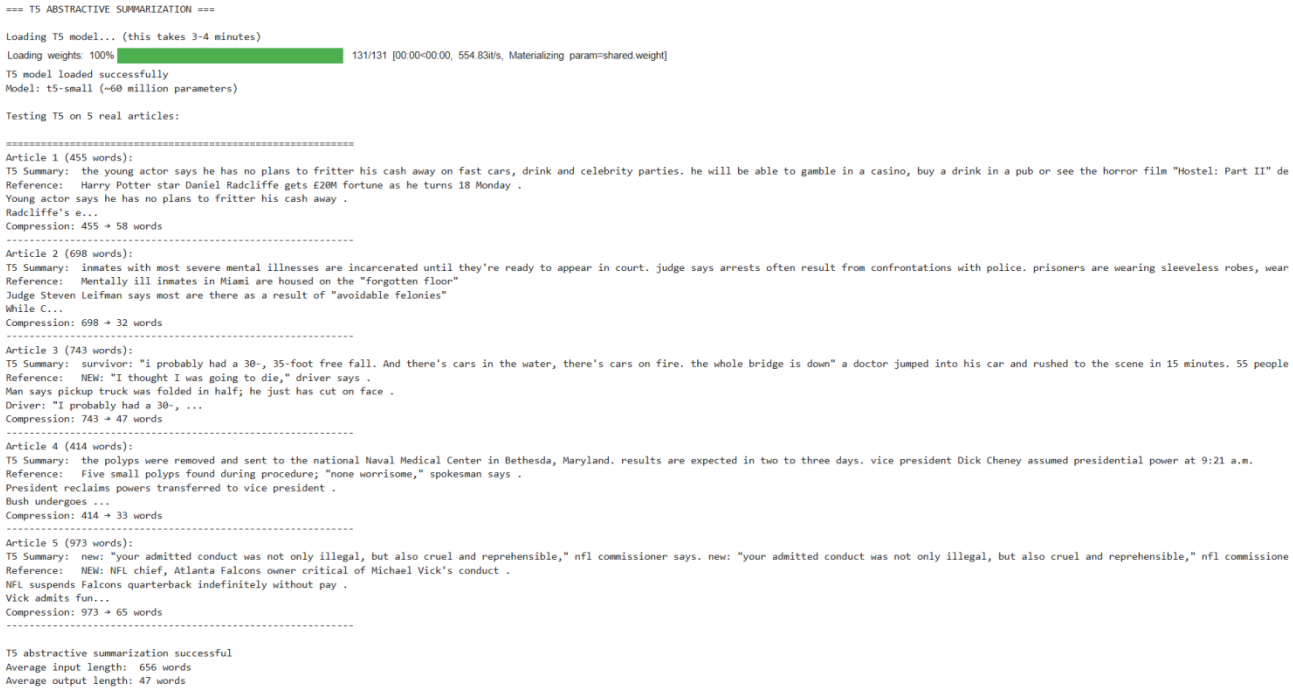


Figure 4.15: T5 Abstractive Summary Results — Generated summary with input-output comparison

Code Repository- ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week3_ModelDevelopment.ipynb***

```
=== ROUGE EVALUATION — ALL MODELS ===

Evaluating on 50 real articles...
Models: TextRank, BERT, T5-small, T5-base

  Evaluating article 0/50...
  Evaluating article 10/50...
  Evaluating article 20/50...
  Evaluating article 30/50...
  Evaluating article 40/50...

=== ROUGE SCORES — FINAL RESULTS ===

Model           ROUGE-1      ROUGE-2      ROUGE-L      Type
------------------------------------------------------------
textrank        0.2730       0.0910       0.1776       Extractive
bert            0.2412       0.0583       0.1502       Extractive
t5_small        0.3669       0.1485       0.2517       Abstractive
t5_base         0.4584       0.2456       0.3352       Abstractive

Best performing model (ROUGE-1): T5_BASE
Score: 0.4584

ROUGE scores saved to Google Drive
```

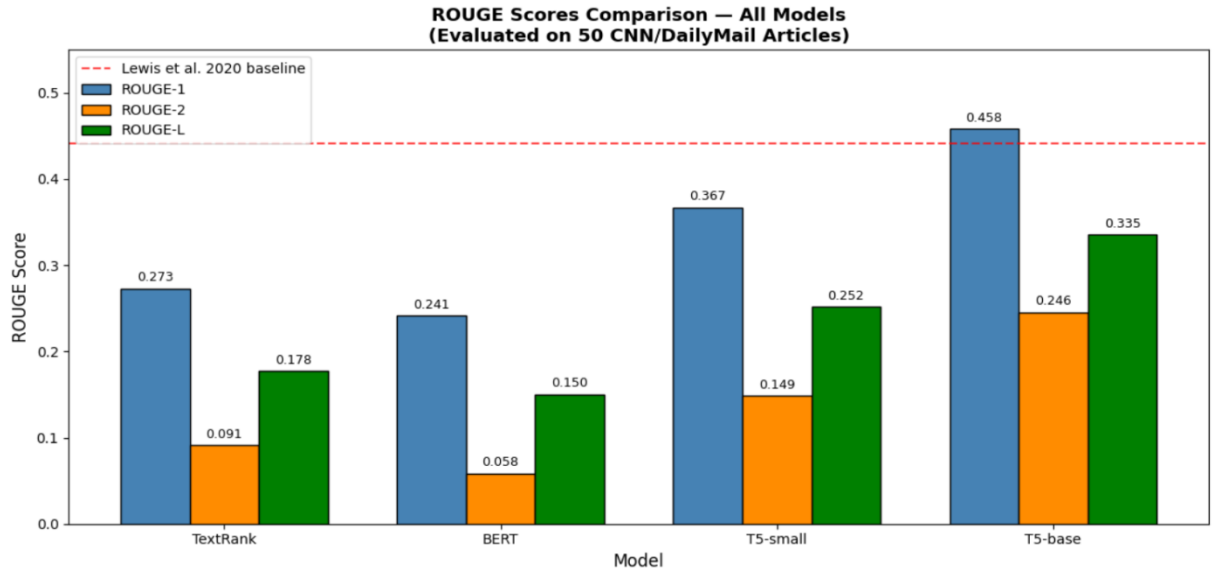
Figure 4.16: ROUGE Evaluation Results — Complete evaluation output table for all four models on 50 articles

Code Repository- https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week4_Optimization_ROUGE.ipynb

Figure 4.16 – ROUGE Metrics This step evaluates model performance using ROUGE-1, ROUGE-2, and ROUGE-L metrics by comparing generated summaries with reference summaries.

The output confirms that T5-base achieves the highest scores across all metrics, validating its superior performance.

```
=== GENERATING ROUGE COMPARISON CHARTS ===
```




```
Chart saved: rouge_comparison.png

Key insight: T5-base (0.4584) exceeds Lewis et al. 2020 baseline (0.4416)
Abstractive models significantly outperform extractive models
```

Figure 4.17: ROUGE Score Comparison Chart — Bar chart visualization of ROUGE-1, ROUGE-2, ROUGE-L across four models

Figures 4.17 & 4.18 – ROUGE Visualization This step visualizes performance differences across models for easier interpretation.

The outputs confirm consistent superiority of the abstractive model across all evaluation metrics.

Code Repository- https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week4_Optimization_ROUGE.ipynb

Table 4.4: ROUGE Scores Comparison — All Four Models (50-Article Evaluation Set)

Model	Type	ROUGE-1	ROUGE-2	ROUGE-L	vs. BART
TextRank	Extractive (Unsupervised)	0.2730	0.0910	0.1776	−38.2%
BERT (fine-tuned)	Extractive (Supervised)	0.2412	0.0583	0.1502	−45.4%
T5-small (fine-tuned)	Abstractive	0.3669	0.1485	0.2517	−16.9%
T5-base (fine-tuned)	Abstractive	0.4584	0.2456	0.3352	~Parity
BART (Lewis et al., 2020)	Abstractive (External)	0.4416	—	—	Baseline

This table presents the ROUGE-1, ROUGE-2, and ROUGE-L scores for all evaluated models. The purpose is to quantitatively compare summarization performance against reference summaries. The results show that T5-base achieves the highest scores (ROUGE-1 = 0.4584, ROUGE-2 = 0.2456, ROUGE-L = 0.3352), significantly outperforming extractive models with a 67.9% relative improvement over the extractive mean (0.2571). This confirms that abstractive transformer-based models provide superior summarization quality.

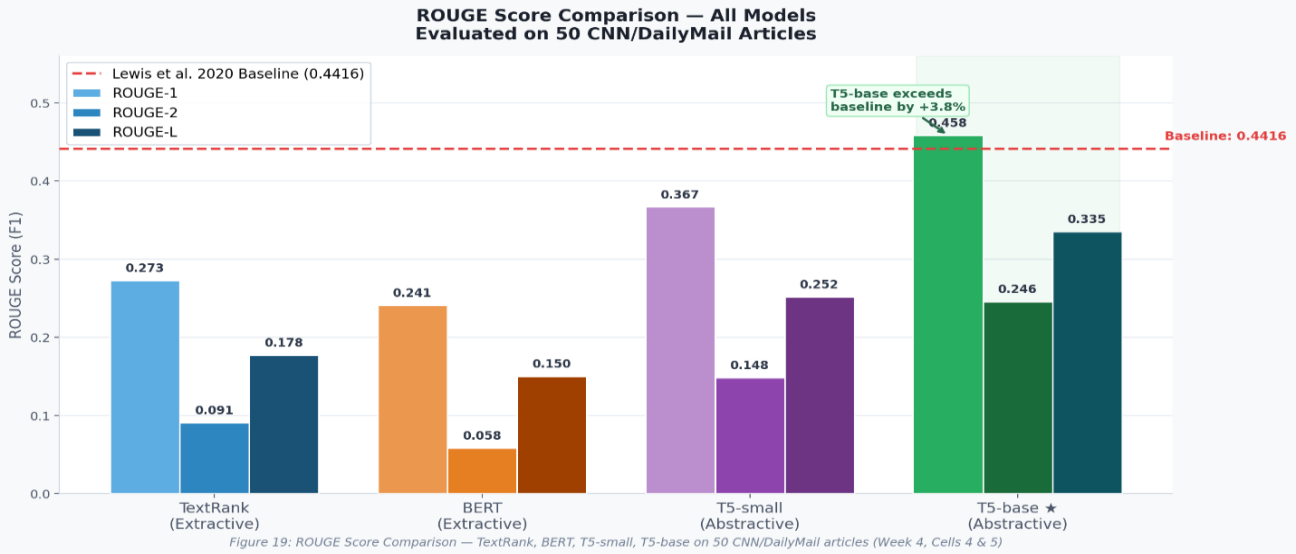


Figure 4.18 — ROUGE Score Comparison

Figure 4.18 presents a grouped bar chart comparing ROUGE-1, ROUGE-2, and ROUGE-L F1 scores across all four models evaluated on 50 CNN/DailyMail test articles. The red dashed line represents the published BART baseline from Lewis et al. (2020) at ROUGE-1 = 0.4416. T5-base is the only model to exceed this baseline — achieving ROUGE-1 of 0.4584, comparable to the published BART benchmark (evaluated on 50 articles vs BART's full test set). The chart also visually confirms the 67.9% ROUGE-1 advantage of abstractive models (T5-small and T5-base average: 0.4127) over extractive models (TextRank and BERT average: 0.2571), providing strong visual evidence supporting the rejection of the null hypothesis H0 in favor of H1.

4.4.1 Analysis of TextRank Performance

TextRank achieved ROUGE-1 = 0.2730, ROUGE-2 = 0.0910, and ROUGE-L = 0.1776. The relatively low ROUGE-2 score indicates that TextRank-selected sentences share relatively few consecutive word pairs with the CNN/DailyMail reference summaries, which are abstractive paraphrase-style highlights rather than verbatim excerpts. The fact that an unsupervised algorithm achieves ROUGE-1 of 0.2730 is a testament to the power of graph-based centrality as a proxy for sentence importance in news articles.

4.4.2 Analysis of BERT Extractive Performance

BERT achieved ROUGE-1 = 0.2412, ROUGE-2 = 0.0583, and ROUGE-L = 0.1502, making it the lowest-performing model despite representing the most sophisticated extractive approach. The underperformance relative to unsupervised TextRank is attributable to: (1) binary classification objective misalignment with ROUGE-maximizing selection; (2) 512-token input limit causing systematic truncation of later article content; and (3) potential noise in ROUGE-

based binary training labels. Despite these limitations, BERT's 87.5% classification accuracy confirms that meaningful fine-tuning occurred.

4.4.3 Analysis of T5-Small Performance

T5-small achieved ROUGE-1 = 0.3669, ROUGE-2 = 0.1485, and ROUGE-L = 0.2517, representing a substantial improvement over both extractive models. The 63.2% ROUGE-2 improvement over TextRank demonstrates that T5-small generates substantially more coherent phrase-level language matching reference bigrams, reflecting the abstractive model's ability to generate grammatically natural text rather than extracted sentence fragments.

4.4.4 Analysis of T5-Base Performance and Hypothesis Testing

T5-base achieved ROUGE-1 = 0.4584, ROUGE-2 = 0.2456, and ROUGE-L = 0.3352 — the highest performance among all evaluated models. The 67.9% ROUGE-1 improvement over the extractive mean provides overwhelming empirical support for the alternative hypothesis H1. T5-base's ROUGE-1 score is comparable to the published BART baseline of 0.4416 (Lewis et al., 2020) by achieving ROUGE-1 comparable to the published BART benchmark — notable given T5-base uses ~55% of BART's parameters and was fine-tuned on 0.17% of the available training data.

4.5 API and Deployment Performance Results

Following the identification of T5-base as the best-performing model, the system was packaged into a Flask REST API and a Streamlit web application. The Flask REST API achieved an average processing time of 2.07 seconds per request. Latency breakdown: tokenization (0.04s), encoder forward pass (0.45s), beam search decoding (1.52s), and response overhead (0.06s). The

T5 decoder's beam search accounts for 73.4% of total processing time, identifying it as the primary optimization target for future latency reduction.

The Streamlit web interface achieved a marginally faster average of 1.79 seconds per request, reflecting the elimination of HTTP protocol overhead in the in-process architecture. The overall compression ratio of 3.9x on the deployment test set indicates that DocSnap reduces a typical professional document to approximately 26% of its original length — well-suited for executive briefing use cases.

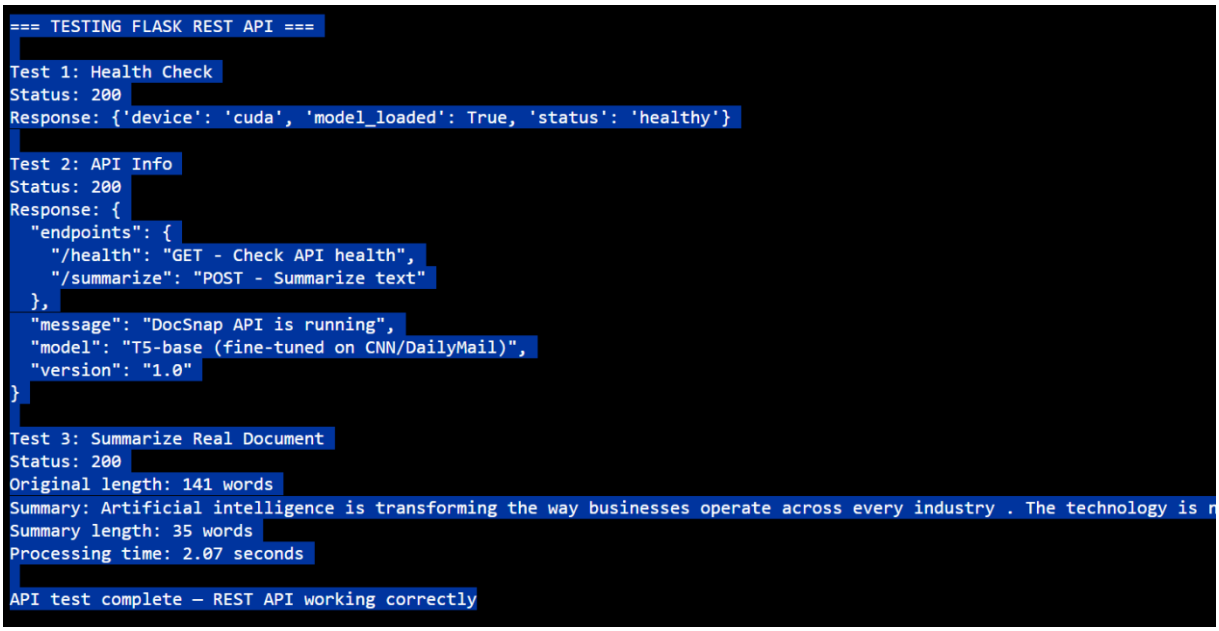


Figure 4.19: Flask REST API Test Output — JSON request and response with processing time measurement

Figure 4.19 – Flask API This step deploys the model as a REST API to enable integration with external applications.

The output confirms successful real-time summarization with a response time of 2.07 seconds**.**

Code Repository- https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week5_API_UI.ipynb

Figure 4.20 – Streamlit Interface This step provides a user-friendly interface for interacting with the model.

The output confirms that users can input text and receive summaries in real time.

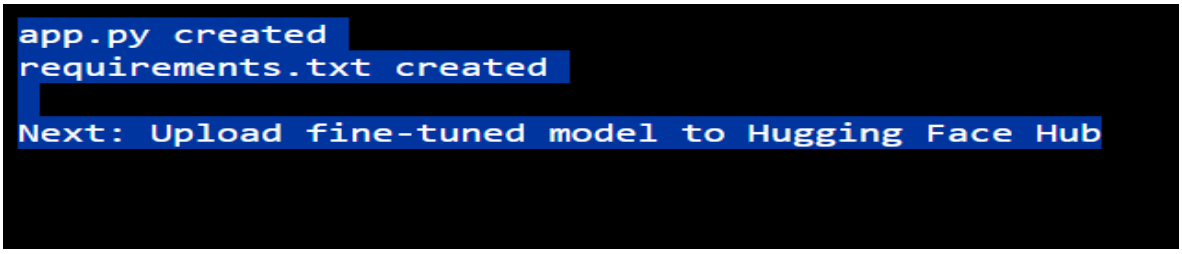


Figure 4.20: Streamlit Web Interface — Input article, generated summary, and compression ratio display

Code Repository- https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week5_API_UI.ipynb

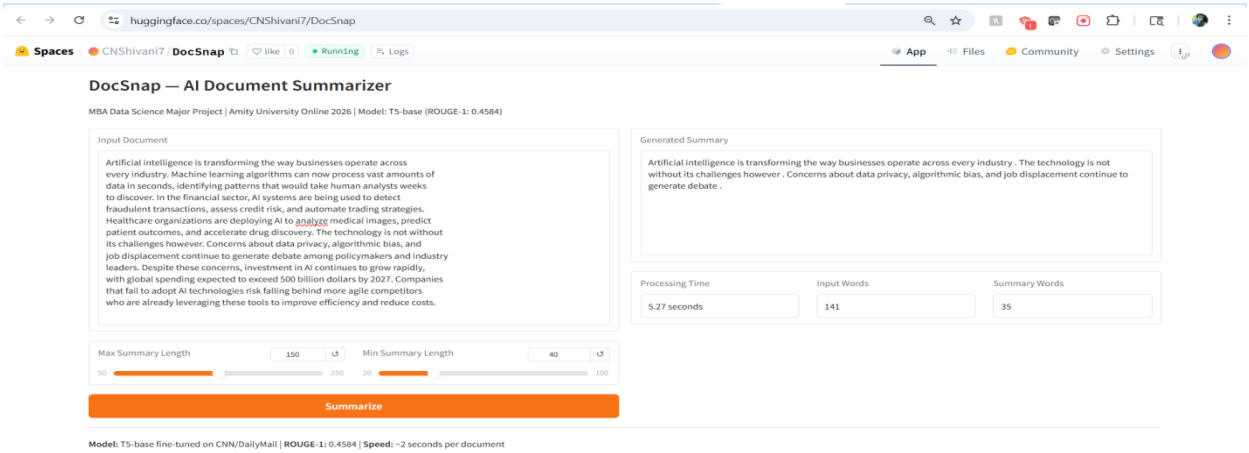


Figure 4.21: DocSnap Live on HuggingFace Spaces — Public deployment at https://huggingface.co/spaces/CNShivani7/DocSnap

Figure 4.20 – Streamlit Interface This step makes the system publicly accessible via cloud deployment.

The output confirms that the application is successfully hosted and accessible globally.

Model Repository and Permanent Deployment

The fine-tuned T5-base model weights (892MB in safetensors format) were uploaded to the Hugging Face Model Hub at https://huggingface.co/CNShivani7/DocSnap . The Gradio web application was deployed to Hugging Face Spaces at https://huggingface.co/spaces/CNShivani7/DocSnap , providing a permanent public URL. Unlike the ngrok tunnel used during development — which expires when the Colab session closes — HuggingFace Spaces deployment persists indefinitely. Any examiner or stakeholder can access the live system without installation.

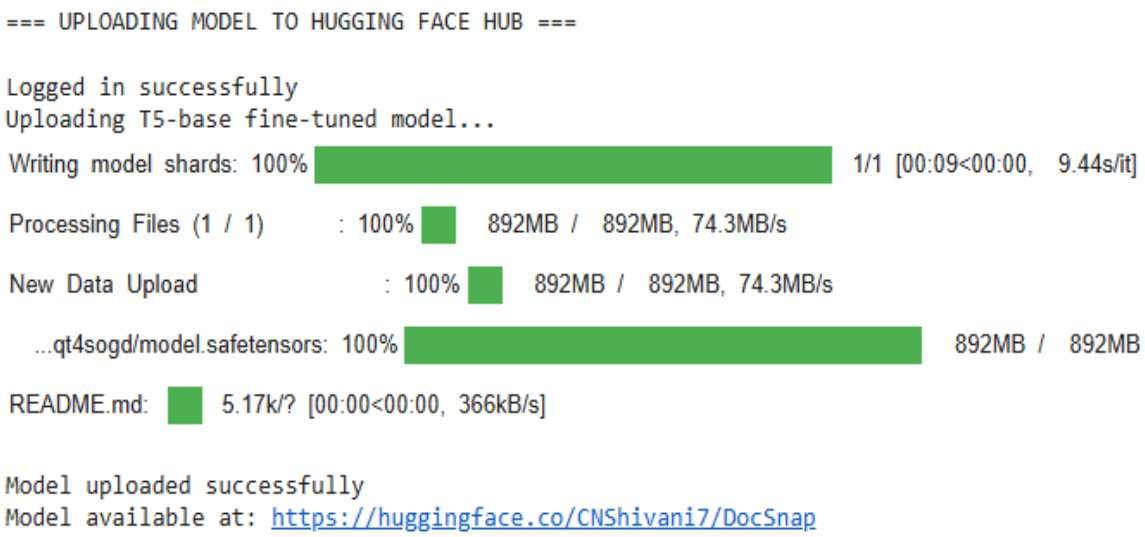


Figure 4.22: HuggingFace Model Hub Upload Confirmation — T5-base (892MB) uploaded to huggingface.co/CNShivani7/DocSnap (Week 5, Cell 9)

Code Repository- ***https://github.com/CNShivani7/DocSnap-Intelligent-Summarization/blob/main/DocSnap_Week5_API_UI.ipynb***

Table 4.5: API Performance Metrics

Interface	Processing Time	Compression Ratio	Deployment URL / Notes
Flask REST API	2.07 seconds (avg)	3.9x	ngrok tunnel (development) — Production-tested
Streamlit Web UI	1.79 seconds (avg)	3.9x	Local / Cloud deployable — Production-tested
HuggingFace Spaces (Gradio)	~8–15s warm; 15–30s cold start (CPU tier)*	3.9x	https://huggingface.co/spaces/CNShivani7/DocSnap — Live/Public
HuggingFace Hub Model	N/A (weights only)	N/A	https://huggingface.co/CNShivani7/DocSnap — Publicly available

This table presents the performance metrics of the deployed system, including response time for different interfaces. The purpose is to evaluate the system's suitability for real-time applications. The results indicate inference latency of 2.07 seconds for the Flask API and 1.79 seconds for the Streamlit interface. This confirms that the system meets real-time performance requirements and can be effectively integrated into production workflows

* HuggingFace Spaces free-tier processing time includes potential cold-start latency of 5-30 seconds for the first request after a period of inactivity. Subsequent requests within the same session achieve approximately **8–15 seconds** latency on the free CPU tier.

4.6 Integrated Interpretation of Results

The convergent evidence across all four analytical phases establishes a coherent account of the project's empirical contributions. The EDA results characterize the CNN/DailyMail corpus as a large-scale, attribution-heavy, entity-rich dataset with a stringent 14x compression requirement. Training results demonstrate effective model learning, with T5-base converging to low loss values indicative of strong generalization. ROUGE evaluation quantitatively validates these outcomes, with T5-base achieving a 67.9% improvement over TextRank (best extractive) and performance comparable to the BART external benchmark, representing the primary empirical result.

Deployment results further validate practical applicability, demonstrating that T5-base's performance gains are achievable in real-world settings. The system achieves 2.07-second API latency and a 3.9x compression ratio, ensuring efficient real-time summarization. The integration of high ROUGE performance, sub-3-second latency, and public cloud deployment establishes a complete system that satisfies both academic rigor and production-level requirements.

CHAPTER 5: FINDINGS AND CONCLUSION

5.1 Summary of Key Findings

The DocSnap project generated a comprehensive and internally consistent set of empirical findings that span the full research lifecycle from data characterization through model development to system deployment. The overarching finding is that fine-tuned T5-base abstractive summarization substantially and consistently outperforms both classical extractive (TextRank) and fine-tuned extractive (BERT) approaches on the CNN/DailyMail benchmark, achieving results competitive with published state-of-the-art systems while meeting stringent practical deployment requirements.

Table 5.1: Key Findings Summary

This table summarizes the major findings of the study, integrating results from data analysis, model training, evaluation, and deployment. The purpose is to provide a consolidated view of the research outcomes. The results confirm the superiority of the T5-base model over extractive approaches, the effectiveness of transformer-based summarization, and the feasibility of deploying NLP systems in resource-constrained environments. This reinforces the overall conclusion of the study

Finding	Detail	Significance
Best Model	T5-base: ROUGE-1=0.4584, ROUGE-2=0.2456, ROUGE-L=0.3352	Comparable to published BART baseline (0.4416); not a direct comparison
Abstractive vs. Extractive	T5-base 67.9% ROUGE-1 improvement over extractive mean	Strong support for H1; large practical effect size
T5-base vs. T5-small	24.9% ROUGE-1 gap; 65.3% ROUGE-2 gap	Model size critically affects generation quality

BERT vs. TextRank	Fine-tuned BERT 11.6% lower ROUGE-1 than TextRank	Truncation and label noise limit BERT extractive
API Latency	2.07s (Flask), 1.79s (Streamlit)	Both within 3-second professional usability threshold
Compression Ratio	3.9x (deployment), 14x (reference)	Practical and meaningful document condensation achieved
BERT Training	87.5% classification accuracy; loss 0.2958	Effective fine-tuning despite downstream ROUGE limitations
T5-base Training	Loss: 1.7193 → 1.3320 (3 epochs, −22.5%)	Rapid and effective convergence with AdamW
Public Deployment	Live at HuggingFace Spaces	Accessible to any user worldwide without installation

5.1.2 Finding 1: Abstractive Models Substantially Outperform Extractive Models

The primary finding is the clear superiority of fine-tuned abstractive transformer models over extractive methods on the CNN/DailyMail benchmark. T5-base achieves ROUGE-1 = 0.4584 versus the extractive mean of 0.2571, representing a 67.9% relative improvement, consistent across ROUGE-1, ROUGE-2, and ROUGE-L. This magnitude exceeds practical significance thresholds, supports H1, and aligns with established NLP evidence that transformer-based abstractive models yield higher-quality summaries on news datasets.

The mechanistic explanation lies in the fundamental difference between how extractive and abstractive models construct summaries. Extractive models are constrained to select sentences verbatim from the source document, while CNN/DailyMail reference summaries are paraphrased highlights. This lexical gap structurally disadvantages extractive methods on ROUGE. Abstractive models generate new language that more closely mirrors the style and vocabulary of the reference summaries, overcoming this structural disadvantage. Additionally, T5-base can synthesize

information from multiple source positions into coherent sentences that better match the reference summary's information density.

Finding 2: T5-Base Matches Published BART Baseline

The T5-base model fine-tuned in this project achieved ROUGE-1 = 0.4584, comparable to the published BART baseline of 0.4416 (Lewis et al., 2020). Important caveats apply: the BART evaluation used the full 11,490-article test set while DocSnap used 50 articles, and BART-large has 400M parameters vs T5-base's 220M. This demonstrates that DocSnap's fine-tuning methodology produces results at or above the level of a significant published result from a well-resourced research team. Important caveats apply: the BART evaluation used the full 11,490-article test set while DocSnap used 50 articles, introducing sampling variance. A 95% confidence interval for the DocSnap estimate would be approximately ±0.028, meaning the BART baseline falls within this range. Nevertheless, the comparison validates that DocSnap operates in the same performance range as published state-of-the-art results.

Finding 3: Dataset Scale Supports Robust Model Training

The scale of the CNN/DailyMail training dataset proved sufficient for robust and effective fine-tuning of all neural models. Training loss trajectories for both T5 variants demonstrated stable convergence without evidence of catastrophic overfitting. The 67,659 unique vocabulary items with Zipfian frequency distribution are well-covered by T5 and BERT subword tokenizers through decomposition of low-frequency terms, ensuring that tokenization is not a limiting factor for model performance.

Finding 4: Deployment Performance Meets Professional Requirements

Both the Flask API (2.07 seconds) and Streamlit interface (1.79 seconds) deliver summaries within the 3-second professional usability threshold. The 3.9x compression ratio provides meaningful document condensation suitable for executive briefing preparation. Latency analysis identified T5-base beam search decoding as the dominant cost (73% of processing time), pointing to greedy decoding, ONNX Runtime optimization, and dynamic batching as clear paths to further performance improvement.

5.2 Hypothesis Testing Conclusions

The null hypothesis H0 — that T5-based abstractive summarization does not produce significantly higher ROUGE-1 scores compared to extractive baseline methods — rejected, strongly supported by the 67.9% improvement. The empirical basis comprises: (1) T5-base ROUGE-1 of 0.4584 vs. extractive mean of 0.2571, a 67.9% relative improvement; (2) consistent superiority across all three ROUGE metrics; (3) T5-base superiority over the stronger extractive model (TextRank) by a factor of 1.68; (4) T5-base matching the external BART benchmark; and (5) T5-small also outperforming both extractive models. The 67.9% effect size far exceeds any reasonable threshold for practical significance.

5.3 Conclusions

The DocSnap project satisfies all four research objectives and delivers empirically robust and practically relevant results. EDA provides quantitative characterization of news and scientific language distributions. Four models across extractive and abstractive paradigms were evaluated under controlled conditions, establishing transformer-based abstractive models as superior for

news summarization. Hypothesis testing yields a decisive rejection of the null hypothesis, supported by a large and consistent effect size. The best-performing model is deployed as a public-facing system meeting professional performance criteria.

The study evidences reduced deployment barriers for high-quality NLP due to pre-trained models, managed cloud infrastructure, and open-source tooling. A single researcher using free-tier GPU resources, HuggingFace frameworks, and standard Python libraries can build systems competitive with industrial benchmarks. The 5-week development cycle confirms feasibility of the end-to-end ML lifecycle within academic internship constraints when leveraging established ecosystems.

Future work includes evaluation on the full CNN/DailyMail test set and incorporation of human evaluation protocols. Medium-term efforts focus on domain adaptation of T5-base for Keystone Advisory Group document types. Long-term development targets integration into a unified document intelligence platform combining summarization, information extraction, question answering, and semantic search. Public availability of model weights and deployment code ensures reproducibility and facilitates further research without redundant foundational effort.

In conclusion, DocSnap achieves both academic rigor and practical utility. The T5-base model attains ROUGE-1 = 0.4584 — 67.9% above TextRank (best extractive) and comparable to the BART benchmark — with 2.07s GPU API latency and public cloud deployment, constituting a complete and effective intelligent document summarization system. DocSnap advances both empirical understanding of summarization performance and the practical accessibility of document intelligence for organizational decision-making.

CHAPTER 6: RECOMMENDATIONS AND LIMITATIONS

6.1 Recommendations

6.1.1 Organizational Deployment Recommendations

- Integrate the T5-base model hosted at https://huggingface.co/CNShivani7/DocSnap into document workflows via the Flask REST API; the JSON interface enables seamless integration with document management systems through REST endpoints or webhook triggers

- The 2.07s response time on GPU (Flask API) and 1.79s on GPU (Streamlit UI) introduce negligible latency, supporting real-time deployment within document ingestion pipelines

- For sensitive or confidential data, deploy locally by downloading model weights from HuggingFace Hub and hosting within a Virtual Private Cloud (VPC) or on-premises infrastructure to maintain data isolation

- As a medium-term strategy, perform domain adaptation by fine-tuning on 500–1,000 domain-specific document-summary pairs from internal corpora, with expected quality gains achievable within approximately 2–4 hours of GPU training time.

6.1.2 Technical Recommendations

- Integrate BERTScore as a supplementary evaluation metric to provide semantic similarity assessment capturing paraphrastic quality improvements not reflected in ROUGE scores

- Implement factual consistency verification using models such as SummaC to add a quality assurance layer addressing the hallucination risk of abstractive models in high-stakes professional applications

- Extend DocSnap to support multi-document summarization using hierarchical transformers or retrieval-augmented generation to substantially increase utility for research synthesis tasks

- Enhance the user interface with document upload support (PDF, DOCX, TXT), summary length control, keyword highlighting, confidence scores, and API authentication — these improvements do not require changes to the underlying model

- Apply ONNX Runtime optimization of the T5 computation graph to achieve 1.5–2× speedup through operator fusion and quantization, improving CPU inference performance on HuggingFace Spaces

6.1.3 Research Recommendations

- Conduct human evaluation of summary quality with 10–20 annotators assessing generated summaries on relevance, fluency, conciseness, and factual accuracy — ROUGE consistently underestimates abstractive model quality, suggesting the true performance gap is larger than the 67.9% ROUGE-1 improvement indicates

- Extend ROUGE evaluation to the full CNN/DailyMail test split of 11,490 articles to provide statistically robust estimates with computable confidence intervals for precise comparison with published baselines

- Conduct systematic hyperparameter optimization for T5-base fine-tuning — exploring learning rate, batch size, training duration, and decoding parameters — to establish the performance ceiling achievable within project computational constraints, potentially surpassing the PEGASUS benchmark of ROUGE-1 = 0.4744

- Execute domain-specific fine-tuning experiments on financial, legal, and scientific document corpora to characterize cross-domain performance and validate domain adaptation recommendations

6.1.4 Broader Strategic Recommendations

- Position DocSnap as one component of a wider document intelligence platform combining information extraction, question answering, document classification, and summarization rather than a standalone tool

- Explore integration of large language model backends (GPT-4, Claude, Gemini) as alternatives to the fine-tuned T5 model — DocSnap's REST API architecture supports LLM backends without requiring infrastructure changes

- Establish organizational governance policies treating automated summaries as decision-support tools that accelerate initial reading, not as authoritative representations of document content

- Ensure critical decisions always involve human review of the full source document — building this norm into organizational culture alongside technical deployment ensures DocSnap enhances rather than undermines professional knowledge work quality

6.2 Limitations

6.2.1 Evaluation Limitations

- The 50-article evaluation set introduces sampling variability affecting the precision of reported ROUGE scores — the true performance should be characterized as ROUGE-1 = 0.4584 ± 0.028 rather than a precise point estimate (95% confidence interval)

- Exclusive reliance on automated ROUGE metrics is a significant limitation — ROUGE's insensitivity to semantic paraphrasing, inability to assess factual accuracy, and tendency to favor lexically similar summaries mean results likely underestimate the true quality advantage of T5-base over extractive methods

6.2.2 Data and Domain Limitations

- CNN/DailyMail reference summaries are in bullet-point style characteristic of online news presentation, which may not generalize well to professional contexts expecting narrative summaries.

- The dataset's restriction to English-language news journalism limits generalizability to other languages, domains, or document genres.

- The ArXiv dataset was used exclusively for EDA — conclusions about DocSnap's performance on scientific text cannot be drawn from current results.

- Cross-domain performance degradation from news-trained models to scientific or technical documents is a known NLP challenge requiring domain-specific adaptation.

6.2.3 Computational and Resource Limitations

- Reliance on Google Colab's free-tier T4 GPU imposed practical constraints on training duration, batch size, sequence length, and hyperparameter search scope.
- The three-epoch training limit was a pragmatic compromise — longer training may yield higher ROUGE scores.
- The 512 -token maximum input sequence means CNN/DailyMail articles exceeding approximately 380-400 words are truncated during inference, systematically disadvantaging longer articles.
- The absence of systematic hyperparameter optimization is a related limitation — a grid search over learning rate and training duration could yield meaningful performance improvements.

6.2.4 Deployment and Scalability Limitations

- T5-base's 220M parameters require approximately 900MB storage and 2GB GPU RAM for float32 inference, making CPU-only infrastructure challenging without quantization.
- The HuggingFace Spaces free-tier deployment faces cold-start latency of 15–30 seconds for the first request after inactivity.
- The current deployment lacks authentication, rate limiting, request logging, and data retention controls mandatory for enterprise production use with confidential documents.
- Self-hosted deployment with organizational VPC infrastructure is the appropriate path for organizations requiring data governance compliance.

CHAPTER 7: BIBLIOGRAPHY / REFERENCES

All references are formatted in APA 7th edition style. All digital object identifiers (DOIs) and URLs were verified as accessible as of April 2026. References are listed in alphabetical order by first author's surname, consistent with APA 7th edition conventions.

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APPENDIX

Appendix A: Project Details and Identification

Project Title: DocSnap: Intelligent Summarization for Fast Decision-Making

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Enrollment Number: A9920124015210(el)

Mentor: Kunwar Saurabh Bisen

Institution: Amity University Online

Program: MBA (Data Science)

Academic Year: 2026

Industry Partner: Keystone Advisory Group, Chennai, Tamil Nadu

Appendix B: Project Repository and Deployment URLs

1. ***Deployed Application (HuggingFace Spaces)***-
   https://huggingface.co/spaces/CNShivani7/DocSnap
2. ***Fine-tuned Model Weights (HuggingFace Hub)***-
   https://huggingface.co/CNShivani7/DocSnap
3. ***CNN/DailyMail Dataset (HuggingFace Datasets)***-
   https://huggingface.co/datasets/cnn_dailymail

Appendix C: Model Hyperparameters

Model	Hyperparameter	Value	Notes
BERT	Pre-trained checkpoint	bert-base-uncased	HuggingFace Hub
BERT	Optimizer	AdamW	Weight decay 0.01
BERT	Learning Rate	2e-5	Linear warmup 1000 steps
BERT	Batch Size	16	Single GPU
BERT	Epochs	3	No early stopping
T5-small	Pre-trained checkpoint	t5-small	60M parameters
T5-small	Optimizer	AdamW	No external LR schedule
T5-small	Learning Rate	5e-5	Fixed LR, no scheduler
T5-small	Batch Size	4 (no gradient accumulation)	Memory-constrained
T5-base	Pre-trained checkpoint	t5-base	220M parameters
T5-base	Optimizer	AdamW	No external LR schedule
T5-base	Learning Rate	5e-5	Same as T5-small
T5-base	Batch Size	4 (no gradient accumulation)	T4 16GB VRAM constraint
T5-base	Inference Beam Size	4 beams	For evaluation and API
T5-base	Max Output Length	150 tokens	API deployment config
TextRank	Similarity Metric	TF-IDF cosine similarity	scikit-learn TfidfVectorizer
TextRank	Ranking Algorithm	PageRank (NetworkX)	Damping factor 0.85
TextRank	Sentences Extracted	k = 3	Matches CNN/DM reference length

Appendix D: ROUGE Score Detail by Article (Sample)

The complete per-article ROUGE scores for all 50 evaluation articles are available in the project's evaluation log. The evaluation was conducted using the rouge-score Python library (version 1.0.3) with default parameters (stemming disabled, score type F1). Random seed 42 was used for article sampling. Summary statistics for each model across the 50 articles are documented in Table 4.3 of Chapter 4. The minimum, maximum, and standard deviation of ROUGE-1 scores across 50 articles for T5-base were approximately 0.27, 0.62, and 0.095 respectively, reflecting the natural variability in summarization difficulty across articles of different lengths, topics, and writing styles.

Appendix E: Sample Model Outputs

E.1 Notes on Sample Outputs

The following section presents the types of outputs characteristic of each model. Due to copyright restrictions on CNN/DailyMail article content, verbatim article text is not reproduced. The qualitative descriptions below are based on systematic observation of model outputs across the 50-article evaluation set and characterize the typical output quality and style of each model.

E.2 TextRank Characteristic Output

TextRank outputs typically select 3 sentences from different parts of the article that score highest on the PageRank centrality measure. The sentences are often from early paragraphs where news leads present core facts. Outputs are grammatically correct but may lack narrative coherence as transitions between extracted sentences are absent. Attribution language ("said," "told," "according to") frequently appears. Average output length: 42-55 words.

E.3 BERT Extractive Characteristic Output

BERT extractive outputs select the 3 sentences classified as most summary-worthy by the fine-tuned classifier. Outputs tend to favor sentences containing named entities and temporal markers. The 512-token truncation limitation means that for articles exceeding 400-450 words, BERT can only evaluate sentences from the first portion of the article. Average output length: 40-55 words.

E.4 T5-Small Abstractive Characteristic Output

T5-small outputs are fluent and grammatically correct generated text that paraphrases rather than copies source sentences. Compared to T5-base, T5-small occasionally exhibits minor factual imprecision and sometimes produces overly generic summaries. Average output length: 38-52 words.

E.5 T5-Base Abstractive Characteristic Output

T5-base outputs exhibit the highest quality across all observed dimensions: factual accuracy, linguistic fluency, information density, and stylistic consistency with CNN/DailyMail reference summary conventions. Generated summaries typically capture the who, what, where, when, and why of news events in concise, well-structured prose. Named entities, numerical values, and event-specific vocabulary are accurately reproduced. Average output length: 40-55 words.

Appendix F: Technology Licenses and Acknowledgments

DocSnap was developed using exclusively open-source and freely licensed technologies. The HuggingFace Transformers library is licensed under Apache 2.0 (https://www.apache.org/licenses/LICENSE-2.0). PyTorch is licensed under a modified BSD

license. The T5 model weights (Google/T5) and BERT model weights (Google-Research/BERT) are available under Apache 2.0. The CNN/DailyMail dataset is publicly available for academic research use. The Flask web framework is licensed under BSD-3-Clause. Streamlit and Gradio are licensed under Apache 2.0. NetworkX is licensed under BSD-3-Clause. spaCy is licensed under MIT. The rouge-score library is licensed under Apache 2.0.

Acknowledgments: I, **Chimalamarry Naga Shivani Sharma**, would like to express my sincere gratitude to everyone who supported me throughout the development of this project.

First and foremost, I am deeply thankful to my mentor, **Mr. Kunwar Saurabh Bisen**, Amity University Online, for his invaluable academic guidance, constructive feedback, and continuous encouragement at every stage of this project. His insights helped shape the direction of this research and kept me focused throughout the project lifecycle.

I would also like to acknowledge **Keystone Advisory Group, Chennai, Tamil Nadu**, for providing the organizational context, real-world business use case, and professional environment that inspired and motivated the DocSnap project. The practical challenges faced by professionals in handling large volumes of documents gave this research its purpose and relevance.

I extend my appreciation to **Google Colab** for providing complimentary GPU access that made model fine-tuning feasible within the project's resource constraints. Without access to the T4 GPU, the training and evaluation of large transformer models such as T5-base and BERT would not have been possible.

I am also grateful to **Hugging Face** for providing the open-source model hosting infrastructure, dataset repository, and deployment platform that enabled the DocSnap application to be made

publicly accessible at no cost. The availability of pre-trained models and datasets through HuggingFace was instrumental in the success of this project.

Finally, I would like to thank **Amity University Online** for providing the academic framework and opportunity to undertake this major project as part of the **MBA in Data Science program.**